

ESWIN

EBC77 Series Single Board Computer User Manual

Rev 2.0

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Notice

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Change History

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1. Safety Notes

1.1. Copyright and Usage Statement

The “EBC77 Series Single Board Computer” (hereinafter referred to as the “SBC”) provided by ESWIN is intended for end-user use only, and shall be used solely for testing and evaluating ESWIN’ s semiconductor components and software applications (the “Intended Use”). No party may resell, redistribute, transfer the product, or use it for commercial gain in any other manner.

Please note that the SBC User Manual may be updated periodically. You may contact ESWIN sales and after-sales support to obtain the latest version.

1.2. Important Safety Information

1.2.1. Warning

- Any power supply used with the SBC must comply with the laws, regulations and standards of the country/region where it is used.
- Operate the SBC only in a well-ventilated environment.
- Use this product only on a flat, non-conductive surface, and never allow contact with conductive materials at any time.
- Connecting incompatible devices to the SBC may damage the single board computer.
- All accessories and peripherals used with the SBC must comply with the standards of the country/region of use and bear the appropriate certification labels to ensure safety requirements are met.
- All cables and connecting devices used with the SBC must be properly insulated in accordance with safety standards and requirements.

1.2.2. Safety Instructions

Observe the requirements below to avoid malfunction and damage to the SBC.

- Do not expose the SBC to water or moisture.
- During operation, do not place the SBC on or near conductive surfaces.
- Do not expose the SBC to high temperatures; use it only at room temperature in a well-ventilated environment as specified in this document.
- Avoid any mechanical or electrical damage to the printed circuit board, interfaces, and integrated circuits on the SBC.
- Avoid touching, lifting, or moving the SBC while it is powered on.

2. Introduction

The EBC77 Series Single Board Computer (hereinafter referred to as SBC) is a high-performance, highly integrated computing platform designed for edge AI computing, multimedia processing, and compact embedded applications. It is built around the latest EIC7700 AI SoC launched by ESWIN (Beijing ESWIN Computing Technology Co., Ltd.), which integrates high-performance SiFive RISC-V CPU cores and a powerful AI acceleration unit.

The SBC features an innovative high-density, single-board design. On its extremely compact board measuring only 85 mm × 56 mm, it integrates a complete system including CPU, memory, storage, and rich multimedia and high-speed interfaces. By eliminating modular separation, this design achieves ultimate compactness, making it an ideal “core” for building space-constrained intelligent terminals, edge boxes, AI cameras, and industrial control equipment.

With its flagship-level hardware configuration, ultra-small form factor, and comprehensive interface support, the SBC can be seamlessly integrated into terminal products with strict space and form-factor requirements. It provides a powerful and reliable hardware foundation for everything from prototype verification to mass deployment.

2.1. Hardware Features

The SBC integrates flagship-grade hardware specifications within an extremely compact form factor, and is designed to provide powerful hardware support for edge AI inference, multimedia processing, and high-performance embedded computing.

Table 1: Key Hardware Specifications of the SBC

Item	Description
Dimensions	85mm x 56mm
CPU	64-bit out-of-order quad-core RISC-V processor
AI Processor	Up to 13.3 TOPS, supports INT8 / INT16 / FP16
Memory	Onboard 8GB/16GB/32GB LPDDR5 @ 6400Mbps
Storage	1x 8MB SPI Flash
	1 × Micro SD card slot
Network	1 × Gigabit Ethernet RJ45 port
	1 × Onboard 802.11ac dual-band Wi-Fi module
Display	1 × Micro HDMI 2.0 output
	1 × 4-lane MIPI DSI TX (compatible with MIPI CSI RX)
Camera	1 × 4-lane MIPI CSI RX (compatible with MIPI DSI TX)
	1x 4-lane MIPI CSI RX
Audio	Via 40-pin expansion header
PCIe	1 × 4-lane PCIe Gen3.0 FPC connector
USB	2 × USB 3.0 Type-A ports (stacked)
	2 × USB 2.0 Type-A ports (stacked)
RTC	1.0mm 2-pin 3V connector
Fan	1.0mm 4-pin 5V connector
Buttons	1x Reset button
	1x Boot select button
Power Supply	USB Type-C PD, recommended ≥ 45W (15V@3A)

Table 1: Key Hardware Specifications of the SBC

Item	Description
Debug	1x Micro-B debug port
Other I/O	I ² C, I ² S, SPI, UART, General-purpose I/O ports (mapped to 40-pin expansion header)
Operating Temperature	-20~60 °C

2.2. Product Components

The SBC adopts a highly integrated single-board design. To help you quickly identify key components on the board and perform subsequent hardware operation and debugging, the following figure numbers and locates the main components, and the table below provides detailed descriptions corresponding to the numbers.

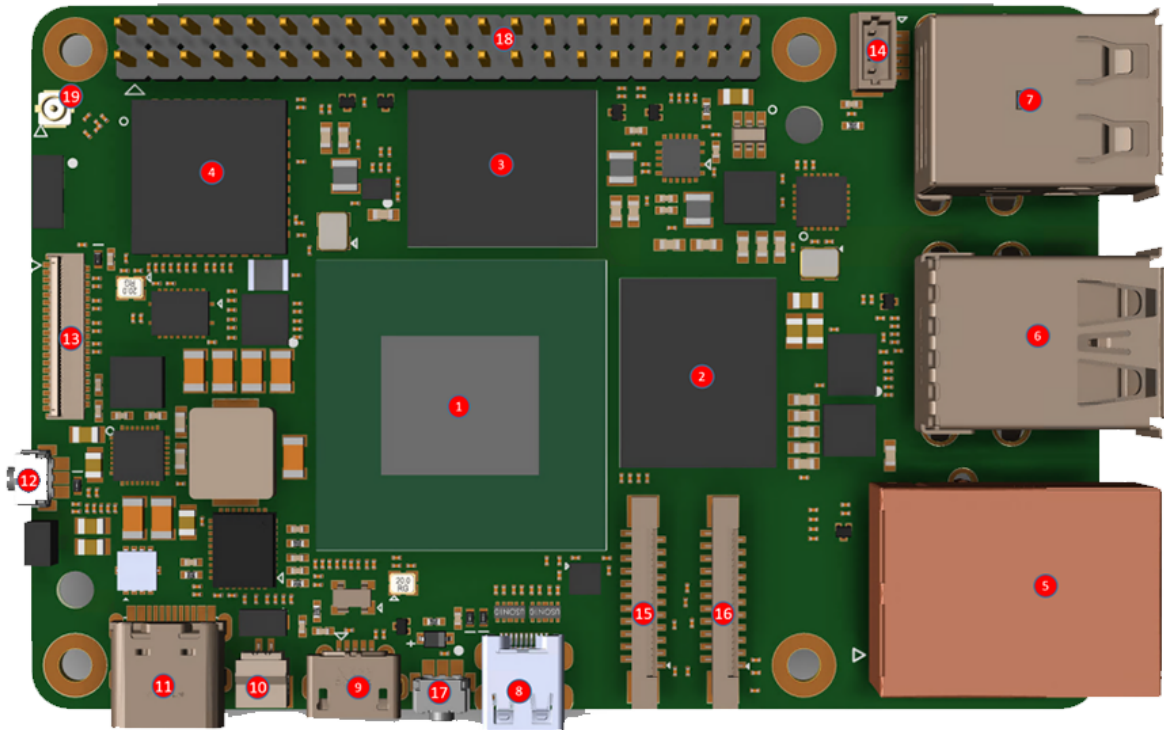


Figure 1: SBC Top View

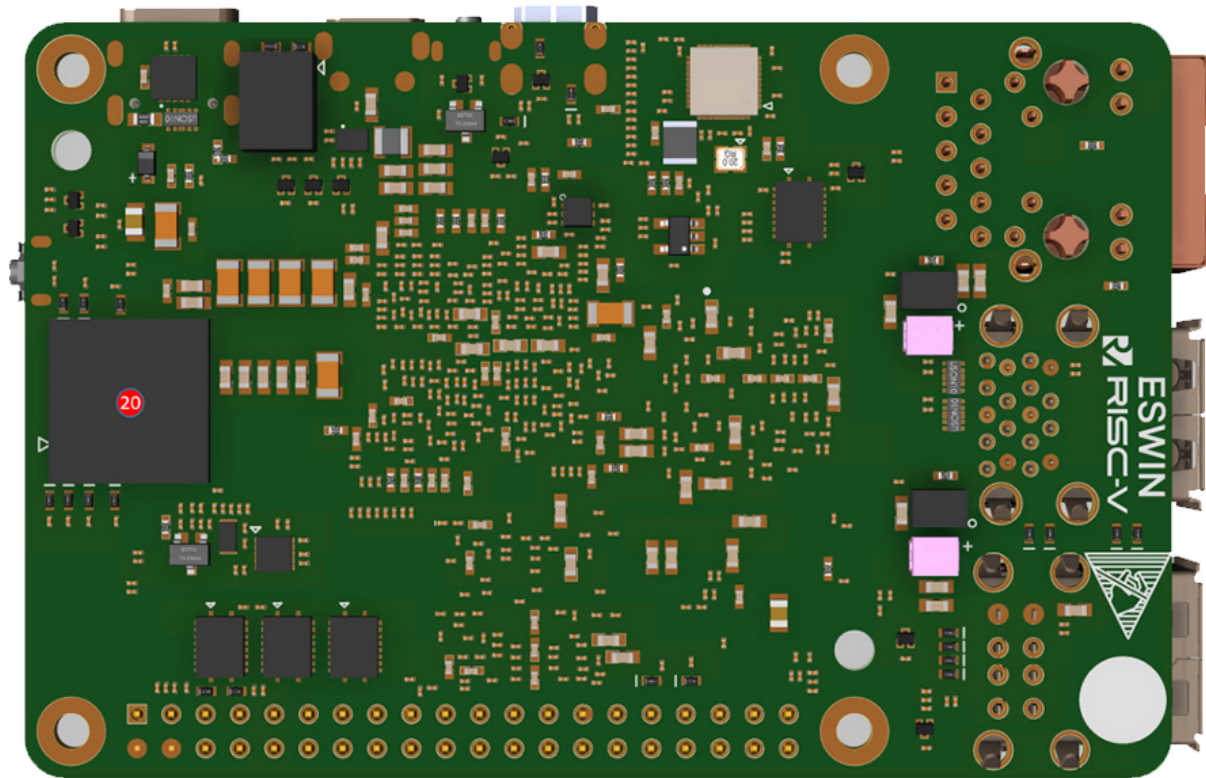


Figure 2: SBC Bottom View

Table 2: SBC Components

No.	Component Name	No.	Component Name
1	EIC7700 SoC	11	USB Type-C PD Power Port
2	LPDDR5 SDRAM	12	System Reset Button (S1)
3	LPDDR5 SDRAM	13	PCIe 3.0 x4 FPC Connector
4	Wi-Fi Module	14	Fan Connector
5	Gigabit Ethernet Port (RJ45)	15	MIPI DSI/CSI Connector (Multiplexed)
6	USB 3.0 Type-A Ports (2x)	16	MIPI CSI Connector
7	USB 2.0 Type-A Ports (2x)	17	Force Recovery Mode Button (S2)
8	Micro HDMI 2.0 Port	18	40-Pin IO Expansion Header
9	Micro USB Debug Serial Port (UART0)	19	Wi-Fi RF Antenna Connector
10	RTC Battery Connector	20	Micro SD Card Slot

2.3. Package Contents

To help you evaluate and use the SBC smoothly, ESWIN provides a full set of supporting accessories. The standard package only includes the bare SBC single-board computer. According to your specific development, testing or integration requirements, you may purchase the following recommended accessories through official ESWIN channels.

Table 3: SBC Optional Accessories

No.	Accessory Name	Quantity	Main Purpose
1	Ethernet Cable	1	For connecting to a wired network via the Gigabit Ethernet port.
2	USB Type-A Data Cable	1	For connecting a PC to the SBC USB port for forced Bootloader recovery mode.
3	Micro USB Data Cable	1	For connecting a PC to the SBC Micro USB debug port for serial port login and debugging.
4	Type-C PD Power Adapter Kit (45W, with dual Type-C cable)	1	Power supply for the SBC, 45W (15V/3A) or higher PD adapter is recommended to ensure stable full-function operation.
5	Micro HDMI to HDMI Cable	1	For connecting to a standard HDMI display for video output.
6	32GB Micro SD (TF) Card	1	The SBC does not have onboard EMMC storage, requiring an SD card for system image storage.
7	SBC Special Heatsink	1	Highly recommended for effective heat dissipation for the EIC7700 SoC, preventing chip damage due to overheating.
8	4-Lane MIPI CSI Camera Module (e.g. IMX219)	1	Video input for computer vision applications.
9	MIPI DSI Display Module	1	For display output via the DSI interface.
10	PCIe Riser Card	1	For PCIe M.2 extension.

2.3.1. Accessory Selection & Purchase Guide

For quick selection based on your usage scenario, please refer to the recommendations below:

Usage Scenario	Essential Core Accessories	Recommended Optional Accessories
Basic Evaluation & Development	4 (PD Power Adapter), 7 (Heatsink), 3 (Debug Cable), 6 (Micro SD Card)	1 (Ethernet Cable), 5 (HDMI Cable), 2 (Flash Cable)
AI & Vision Application Development	Basic evaluation accessories + 8 (CSI Camera Module)	
Display Terminal Application	Basic evaluation accessories + 9 (DSI Display Module)	
Storage & High-Speed Extension	Basic evaluation accessories + 10 (PCIe Riser Card)	

Tip

How to Obtain: All accessories listed above are optional. You may consult and purchase them through official sales or technical support channels of ESWIN (Eswin Computing). Please verify that the accessory models are compatible with your SBC version before purchase.

2.4. Software & Documentation

The SBC provides a complete software and hardware support ecosystem. In addition to this User Manual, which guides hardware operation and basic flashing, for more advanced work such as system

customization, driver development, and AI application deployment, please refer to the following resources:

- [BSP Quick Start Guide](#): Guides on setting up the cross-compilation environment, customizing the Linux kernel, building the root filesystem, and compiling the full SDK.
- [ESSDK Quick Start Guide](#): Detailed introduction to the structure and usage of the ESWIN SDK, as well as AI model quantization, compilation, and on-board deployment & inference.

You may obtain the latest documentation, tools, and images through official ESWIN (Eswin Computing) technical support channels.

3. Design Block Diagram

3.1. EIC7700 Chip Block Diagram

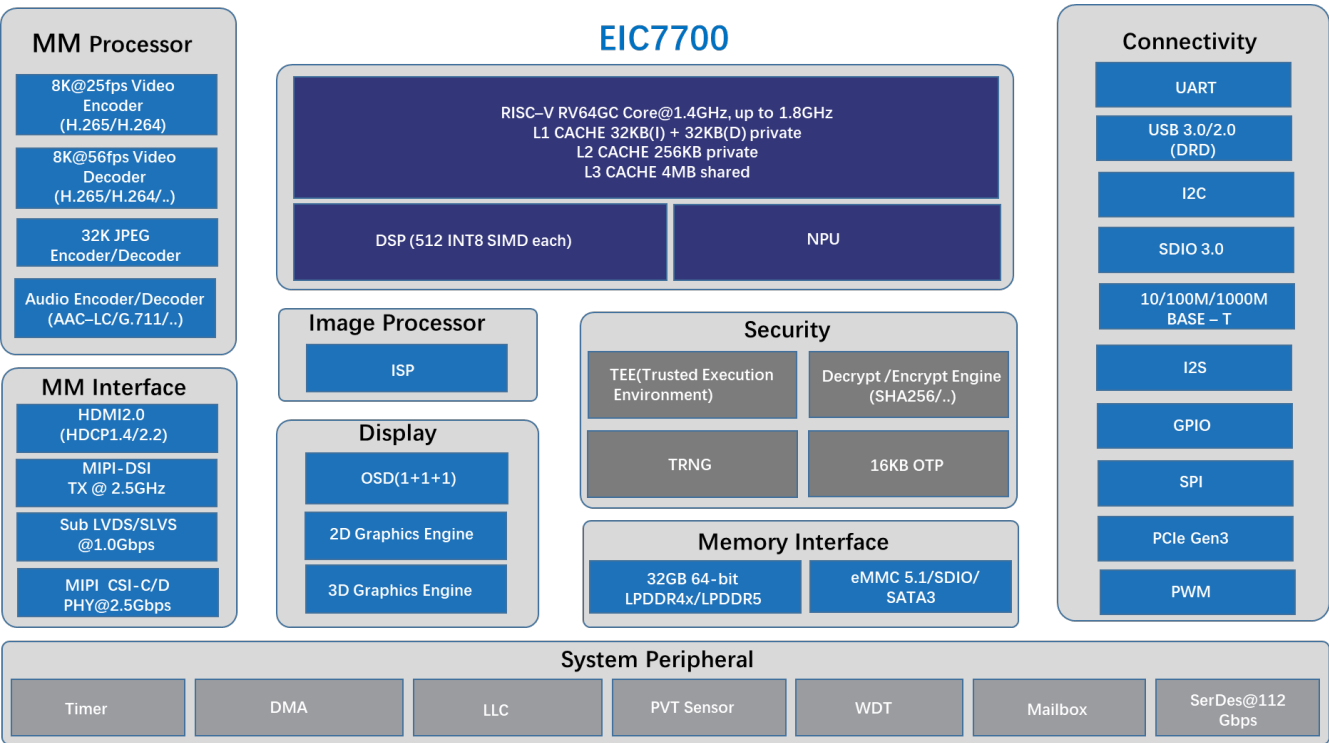


Figure 3: EIC7700 Chip Block Diagram

3.2. System Block Diagram

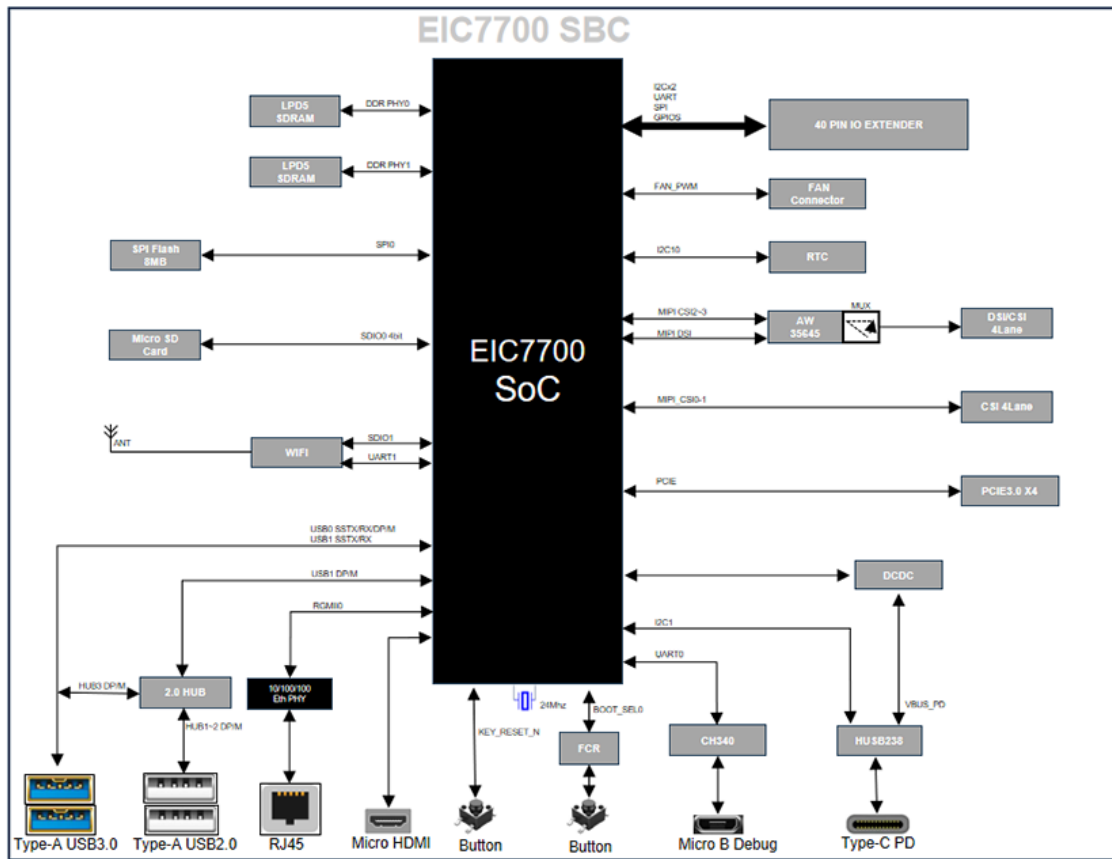


Figure 4: SBC System Block Diagram

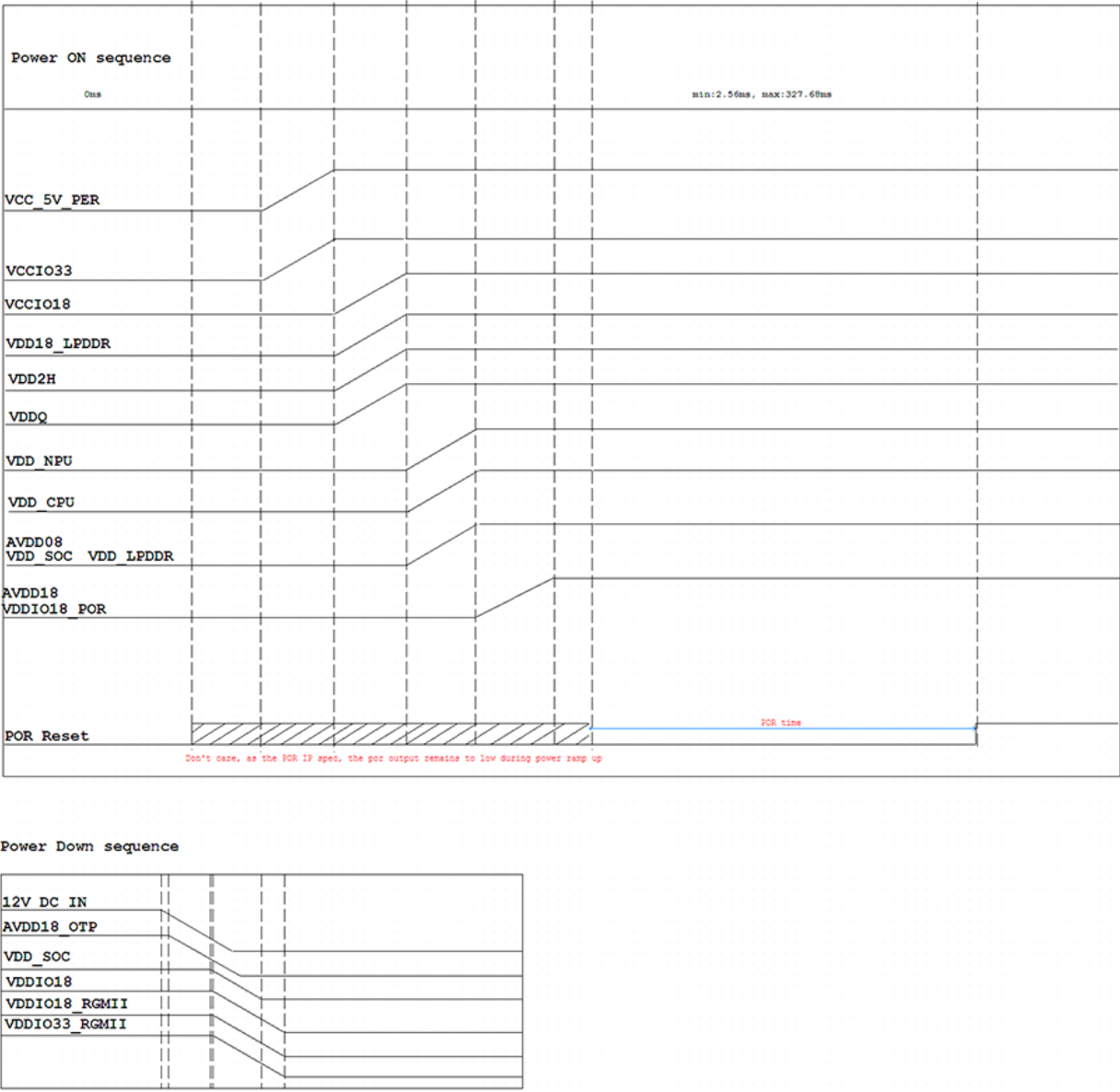


Figure 6: SBC Power-On Sequence

4. Hardware Introduction and User Guide

4.1. Preparations

Before applying power to the SBC, please complete all preparations in this section to ensure safe and proper operation of the device.

4.1.1. Heatsink and Fan

To prevent overheating and damage to the core chip, a heatsink must be installed on the SBC before first power-on.

Warning

Powering on the board without a heatsink installed may cause permanent thermal damage to the EIC7700 SoC.

Steps:

1. Align the dedicated heatsink with the SoC and install it securely.
2. Connect the 5V fan to the onboard 4-pin fan header.

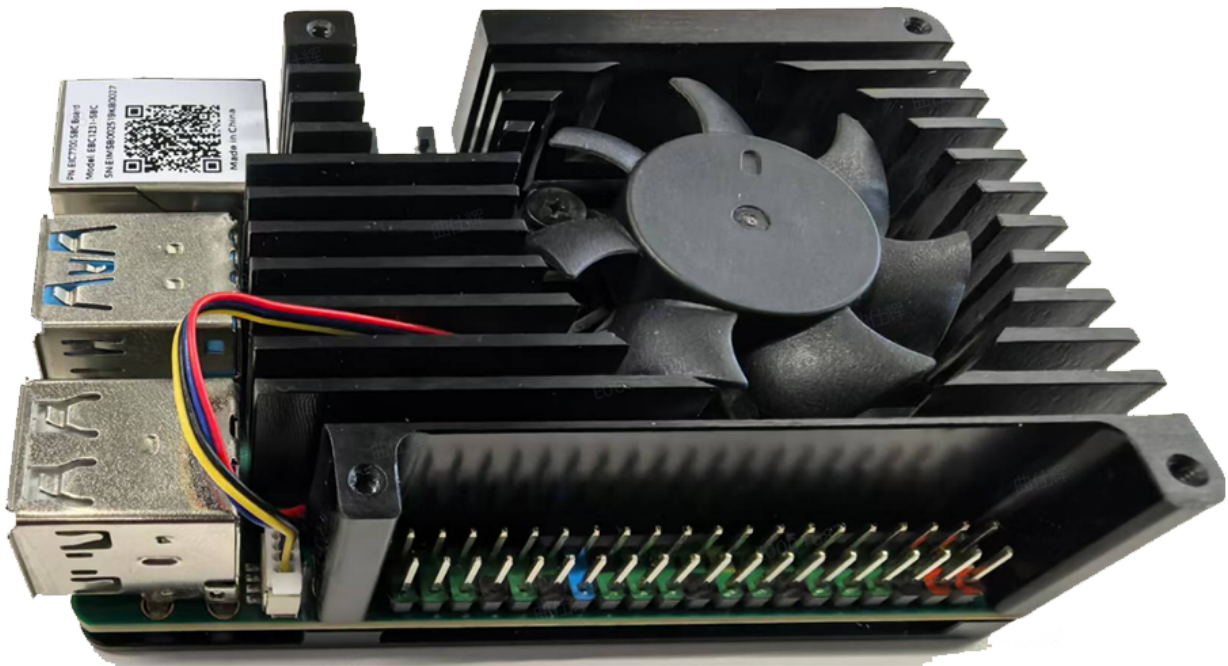


Figure 7: heatsink

4.1.2. Peripheral Connections

To facilitate subsequent debugging, operation, and function verification, it is recommended to connect the following common peripherals before first power-on.

Peripheral	Connection Interface	Main Purpose
Power Supply	USB Type-C PD Port	Connect a 45W (15V/3A) or higher-power PD adapter. This is the only power supply method for the SBC.
System Media	Micro SD Card Slot	Insert the Micro SD card with pre-burned Ubuntu system. The SBC has no onboard eMMC; the system runs from the SD card.
Debug & Command Line	Micro USB Port	Connect to your PC for serial debugging, viewing system boot information and console login.
Display Output	Micro HDMI Port	Connect a monitor for viewing the graphical desktop.
Input Devices	Any USB Type-A Port	Connect keyboard and mouse for operating GUI or command line after boot.
Network	Gigabit Ethernet (RJ45) Port	Connect network cable to access the local area network.

4.1.3. Power On

The SBC supports power supply only via the USB Type-C PD 3.0 interface.

Warning

1. The SBC maximum supported voltage is 15V. Using a voltage higher than 15V is strictly prohibited and will result in permanent damage to the device.
2. It is strongly recommended to use a PD adapter with a rated power $\geq 45\text{W}$ (15V/3A) to ensure stable operation under full system load.
3. The power adapter negotiates the output voltage (15V, 12V, 9V, or 5V) with the SBC according to the PD protocol. Relying on the 5V level is not recommended, as voltage drop on some adapters may cause boot failure.

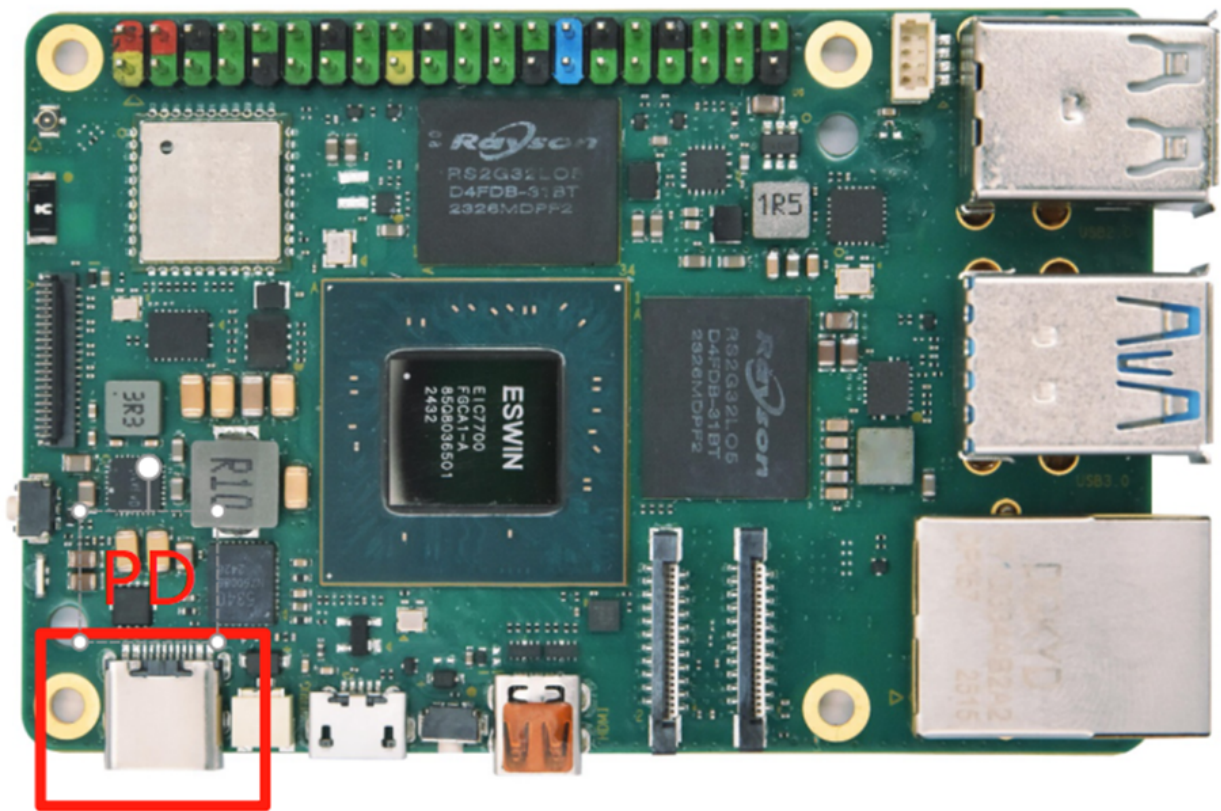


Figure 8: PD Power Supply

Steps:

1. Connect the qualified PD power adapter to the SBC Type-C PD power port using a Type-C cable.
2. Observe the onboard RGB LED indicator. If it lights up green continuously, the PD protocol negotiation is successful and the board is powered on properly.

4.2. Power On/Off, Boot Mode and Status Identification

After completing hardware preparation, you can learn how to power on and operate the SBC through this section, and recognize system status via buttons and indicators.

4.2.1. Button Functions and Operation

The SBC provides two physical buttons for system reset and entering recovery mode.

Button Name	Main Function	Operation Method
Reset Button (S1)	Hardware reset of the entire system.	Short press: Triggers an immediate SoC hardware reboot, equivalent to the “Restart” button on a PC.
Recovery Button (S2)	Enter forced recovery mode.	Used for system recovery when the bootloader is corrupted. Multiple trigger methods are available, see Forced Recovery Mode below.

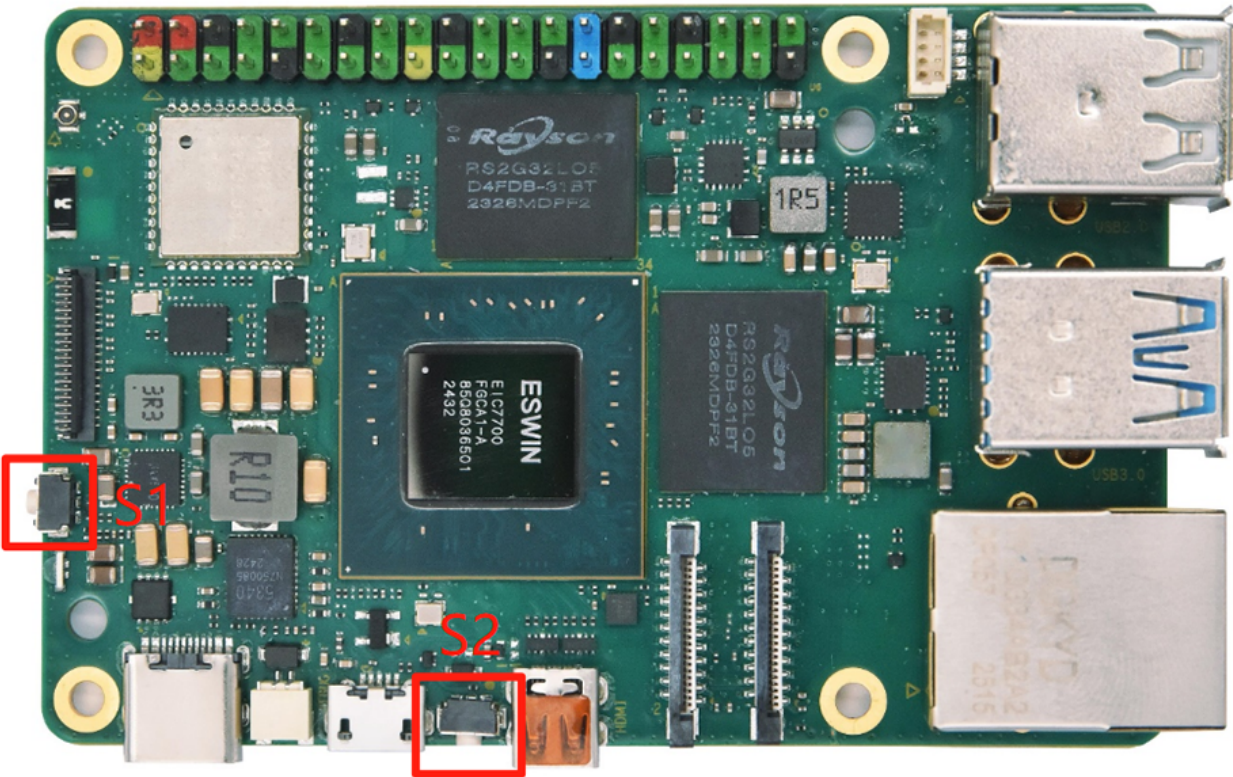


Figure 9: Button

4.2.2. Status Indicator (LED) Interpretation

The SBC is equipped with a programmable RGB LED indicator to indicate core system status.

Indicator Color	Status & Meaning
Green Solid On	Power Normal: PD power supply has been successfully connected, and core system power is stable. This is the default state after power-on.
Blue/Red	User-Defined: Can be programmatically controlled by user applications via GPIO to indicate specific application statuses (e.g., system busy, error alerts, etc.).

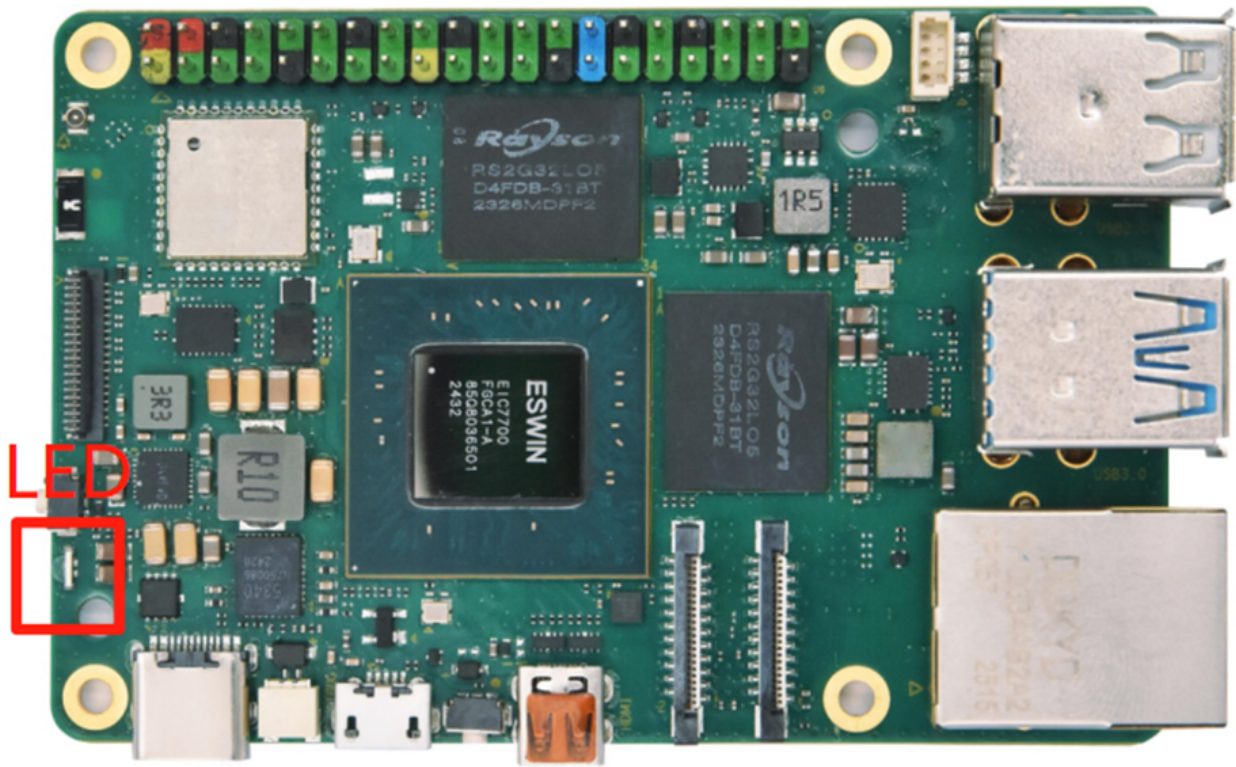


Figure 10: SBC LED Location

4.2.3. Power On/Off Operation Procedure

The SBC has no physical power switch. Its power-on and power-off procedures are as follows:

1. **Power On / Boot:** After connecting a compliant PD power supply, the system will automatically power on and boot. Observe that the RGB LED stays solid green, and check the boot progress via the serial port or HDMI monitor.
2. **Normal Shutdown:** In the running operating system (e.g., Ubuntu), execute the shutdown command (e.g., enter `sudo poweroff` in the terminal).
3. **Forced Power Off:** Directly unplug the Type-C power cable. Note: This operation may cause data loss and is only recommended when the system is completely unresponsive.

4.2.4. Serial Port Debugging

The SBC development board provides UART0 serial port debugging via the Micro-B interface.

1. **Driver Installation:** On a Windows PC, install the CH340/CH341 USB-to-serial driver first. The driver can be downloaded from: https://www.wch.cn/downloads/ch341ser_exe.html
2. **Hardware Connection:** Connect the Micro-B interface of the SBC to the USB port of the PC using a Micro USB cable.
3. **Software Configuration:** Use a serial terminal tool (such as MobaXterm, PuTTY, SecureCRT, etc.) on the PC to connect to the corresponding COM port, and set the baud rate to 115200 8N1.

4.3. Interface Details

The SBC integrates a rich set of interfaces in a compact form factor. This chapter will detail the location, functions, and connection considerations of all interfaces.

4.3.1. USB Interface

The SBC is equipped with a total of 5 USB interfaces, whose layout and functions are as follows:

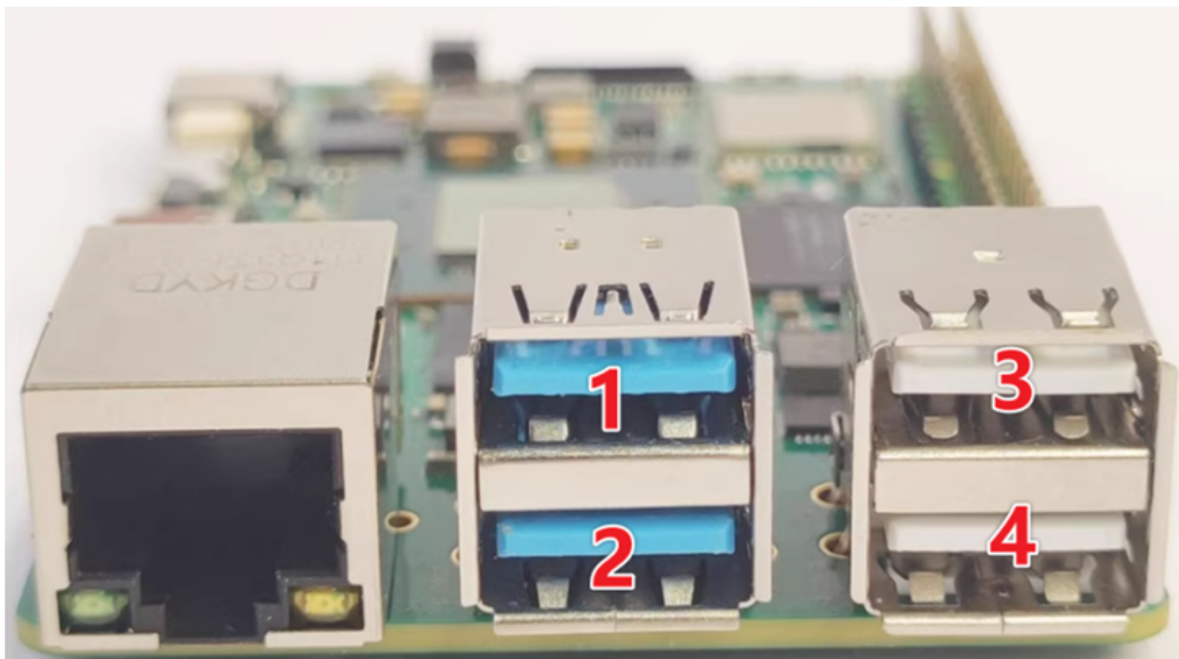


Figure 11: USB Interface

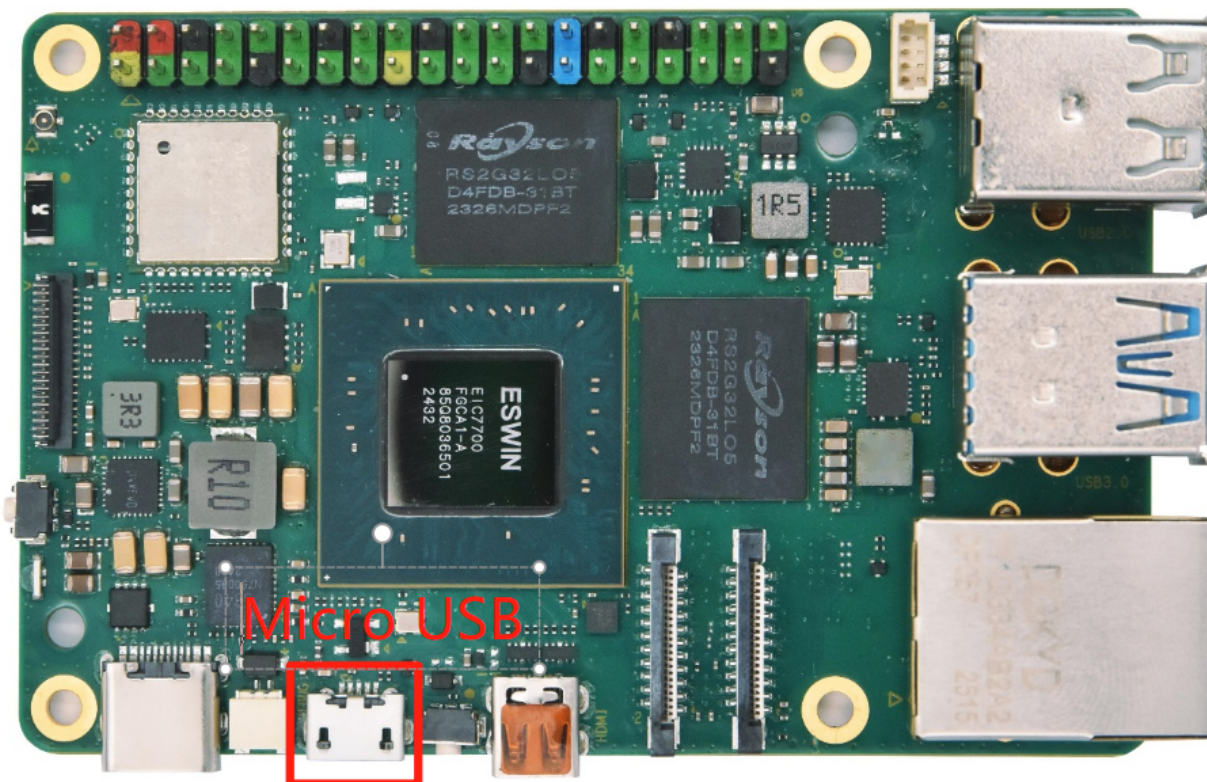


Figure 12: Micro USB Interface

Table 4: SBC USB Interface Function Description

Interface Location	Type	Function Description
Port 1	USB 3.2 Gen1 Type-A	Connects to USB 3.0 devices. Shares current with Port 4. <i>Note: This interface is unavailable in U-Boot.</i>
Port 2	USB 3.2 Gen1 Type-A	Special function port: 1. Serves as the software download interface in forced recovery mode. 2. Can act as Device mode for Fastboot flashing in U-Boot. Shares current with Port 3.
Port 3	USB 2.0 Type-A	Connects to USB 2.0 devices. Shares current with Port 2.
Port 4	USB 2.0 Type-A	Connects to USB 2.0 devices. Shares current with Port 1.
Micro USB	USB 2.0 Micro-B	Debug serial port (UART0): Connects to a PC for serial port debugging and login.

4.3.2. Display Interfaces

The SBC supports two display output options, which cannot operate simultaneously and require software-level configuration.

- **Micro HDMI:** Supports HDMI 2.0/1.4b, with a maximum output resolution of 4K@60Hz.
- **MIPI DSI FPC Interface:** 4-lane MIPI DSI TX, supporting up to 1080P@60Hz. This interface is

hardware-multiplexed with the MIPI CSI23 interface; see [MIPI Display Interface \(DSI\)](#) for details.

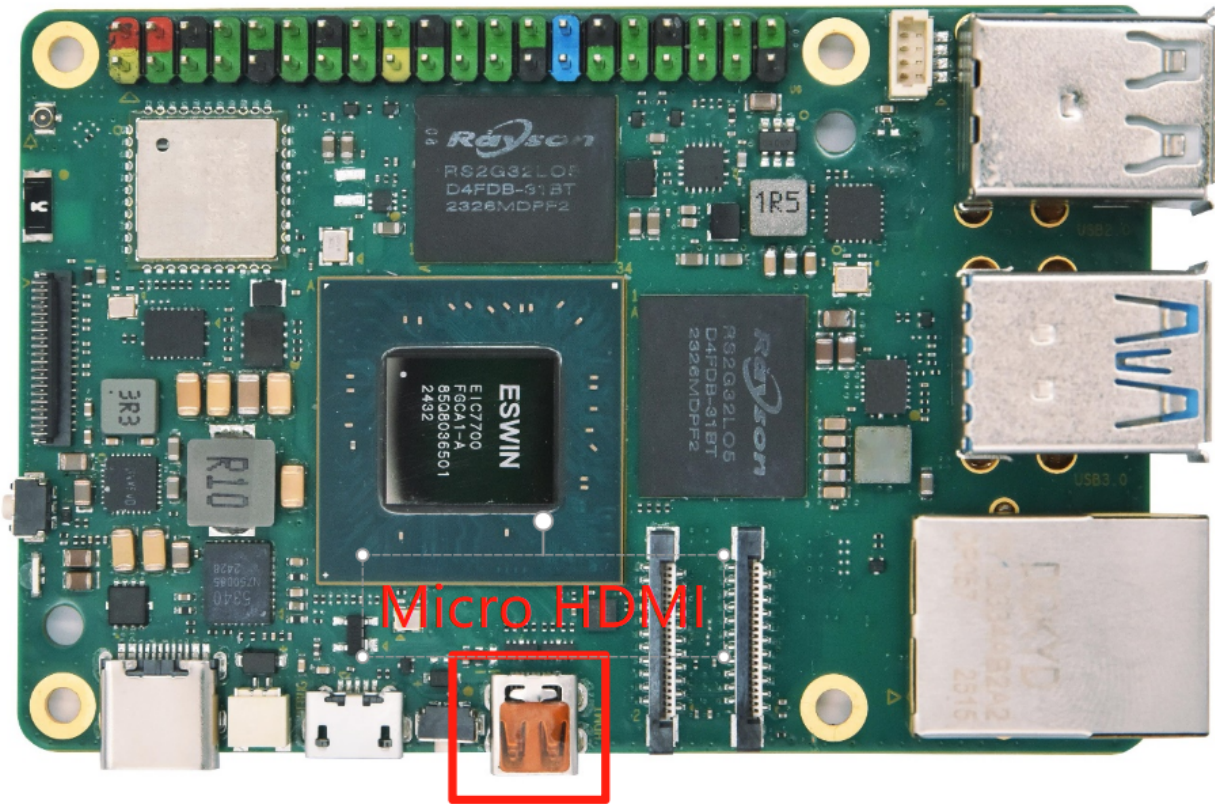
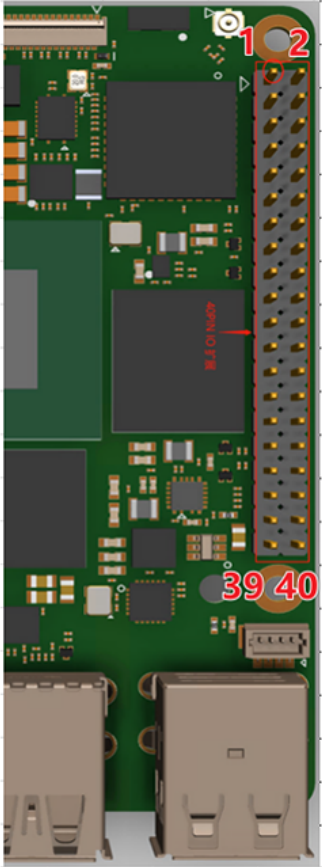


Figure 13: Micro HDMI

4.3.3. 40-Pin GPIO Expansion Interface

This pin header brings out a rich set of system signals for users to expand custom hardware functions, including multiple I²C, I²S, SPI, UART, PWM, and GPIO channels.



3.3V	1	2	5V
I2C5_SDA	3	4	5V
I2C5_SCL	5	6	GND
GPIO_022	7	8	UART3_TXD
GND	9	10	UART3_RXD
GPIO_009	11	12	GPIO_030
GPIO_001	13	14	GND
GPIO_008	15	16	GPIO_010
3.3V	17	18	GPIO_004
MOSI	19	20	GND
MISO	21	22	GPIO_003
SCLK	23	24	CE0
GND	25	26	CE1
I2C4_SDA	27	28	I2C4_SCL
GPIO_007	29	30	GND
GPIO_062	31	32	GPIO_039
GPIO_063	33	34	GND
GPIO_031	35	36	GPIO_040
GPIO_022	37	38	GPIO_032
GND	39	40	GPIO_033

Figure 14: 40-Pin GPIO Expansion Interface

Table 5: SBC 40-Pin Expansion Interface Pin Definition

Pin No.	Name	Type	Voltage	Default Function	Description
1	3.3V	Power	3.3V		3.3V Power Output
2	5V	Power	5.0V		5V Power Output
3	SDA.1	I/O	3.3V	I2C Data Line	I2C5_SDA, can be multiplexed as GPIO55
4	5V	Power	5.0V		5V Power Output
5	SCL.1	Output	3.3V	I2C Clock Line	I2C5_SCL, can be multiplexed as GPIO54
6	GND	Ground			Signal Ground
7	GPIO_022	I/O	3.3V	General Purpose I/O	I2S_MCLK, can be multiplexed as GPIO22
8	TXD	Output	3.3V	UART Send Data	UART3_TX, can be multiplexed as GPIO92
9	GND	Ground			Signal Ground
10	RXD	Input	3.3V	UART Receive Data	UART3_RX, can be multiplexed as GPIO93
11	GPIO_009	I/O	3.3V	General Purpose I/O	JTAG1_TDI, can be multiplexed as GPIO09

Table 5: SBC 40-Pin Expansion Interface Pin Definition

Pin No.	Name	Type	Voltage	Default Function	Description
12	GPIO_030	I/O	3.3V	General Purpose I/O	I2S1_BCLK, can be multiplexed as GPIO30
13	GPIO_001	I/O	3.3V	General Purpose I/O	JTAG0_TCK, can be multiplexed as SPI2_CLK or GPIO1
14	GND	Ground			Signal Ground
15	GPIO_008	I/O	3.3V	General Purpose I/O	JTAG1_TMS, can be multiplexed as GPIO8
16	GPIO_010	I/O	3.3V	General Purpose I/O	JTAG1_TDO, can be multiplexed as GPIO10
17	3.3V	Power	3.3V		3.3V Power Output
18	GPIO_004	I/O	3.3V	General Purpose I/O	JTAG0_TDO, can be multiplexed as SPI2_D2 or GPIO4
19	MOSI	Output	3.3V	SPI Master Out Slave In	SPI1_D0, can be multiplexed as GPIO37
20	GND	Ground			Signal Ground
21	MISO	Input	3.3V	SPI Master In Slave Out	SPI1_D1, can be multiplexed as GPIO38
22	GPIO_003	I/O	3.3V	General Purpose I/O	JTAG0_TDI, can be multiplexed as SPI2_D1 or GPIO3
23	SCLK	Output	3.3V	SPI Clock	SPI1_CLK, can be multiplexed as GPIO36
24	CE0	Output	3.3V	SPI Chip Select 0	SPI1_CS0_N, can be multiplexed as GPIO35
25	GND	Ground			Signal Ground
26	CE1	Output	3.3V	SPI Chip Select 1	SPI1_CS1_N, can be multiplexed as GPIO41 or PWM2
27	SDA.0	I/O	3.3V	I2C Data Line	I2C4_SDA, can be multiplexed as GPIO53
28	SCL.0	Output	3.3V	I2C Clock Line	I2C4_SCL, can be multiplexed as GPIO52
29	GPIO_007	I/O	3.3V	General Purpose I/O	JTAG1_TCK, can be multiplexed as GPIO7
30	GND	Ground			Signal Ground
31	GPIO_062	I/O	3.3V	General Purpose I/O	Dedicated GPIO62
32	GPIO_039	I/O	3.3V	General Purpose I/O	SPI1_D2, can be multiplexed as GPIO39
33	GPIO_063	I/O	3.3V	General Purpose I/O	Dedicated GPIO63
34	GND	Ground			Signal Ground
35	GPIO_031	I/O	3.3V	General Purpose I/O	I2S1_WCLK, can be multiplexed as GPIO31
36	GPIO_040	I/O	3.3V	General Purpose I/O	SPI1_D3, can be multiplexed as GPIO40 or PWM1

Table 5: SBC 40-Pin Expansion Interface Pin Definition

Pin No.	Name	Type	Voltage	Default Function	Description
37	GPIO_002	I/O	3.3V	General Purpose I/O	JTAG0_TMS, can be multiplexed as SPI2_D0 or GPIO2
38	GPIO_032	I/O	3.3V	General Purpose I/O	I2S1_SDI, can be multiplexed as GPIO32
39	GND	Ground			Signal Ground
40	GPIO_033	I/O	3.3V	General Purpose I/O	I2S1_SDO, can be multiplexed as GPIO33

4.3.4. PCIe Expansion Interface

- **Interface Type:** FPC connector (0.3mm pitch, part number HC-FPC-03-10-39RLTAG)
- **Specification:** PCIe Gen3 x4
- **Operating Mode:** Supports RC (Root Complex) and EP (Endpoint) modes.

Warning

1. **Gold Finger Orientation:** When connecting the FPC cable, the gold fingers must face downward.
2. **Clock:** An external 100MHz differential clock input is required for both RC and EP modes.
3. **Power Supply:** When the SBC operates as RC, it can provide 5V power to the daughter card via FPC; when operating as EP, the 5V pin on the FPC must NOT be connected.

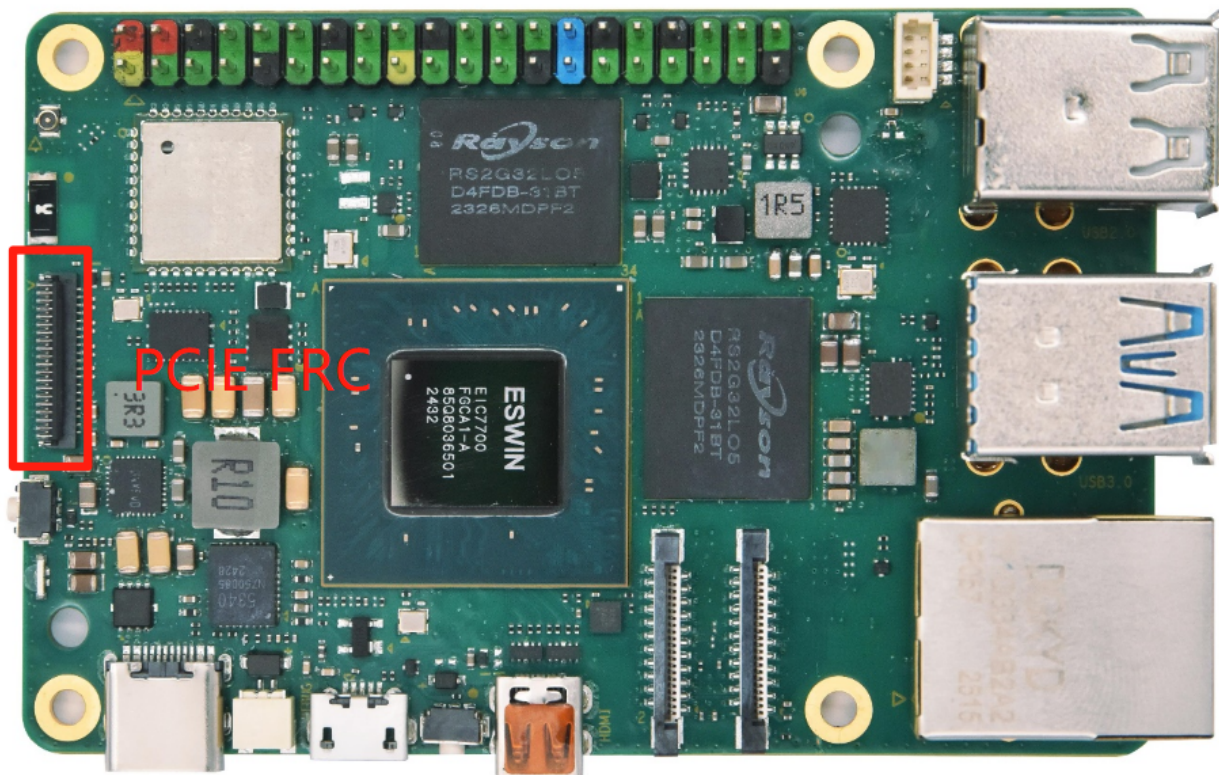


Figure 15: PCIE FPC

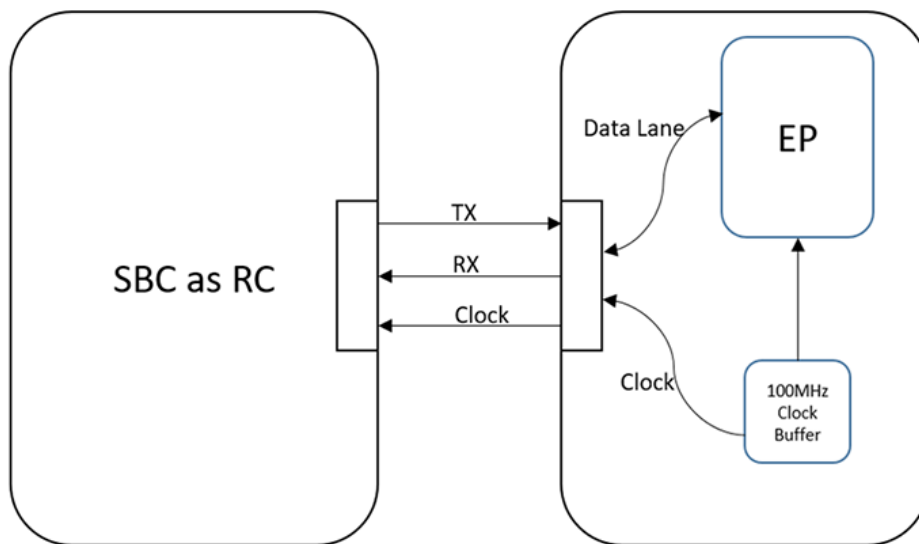
PCIE RC Connection Diagram:

Figure 16: PCIE RC

PCIE EP Connection Diagram:

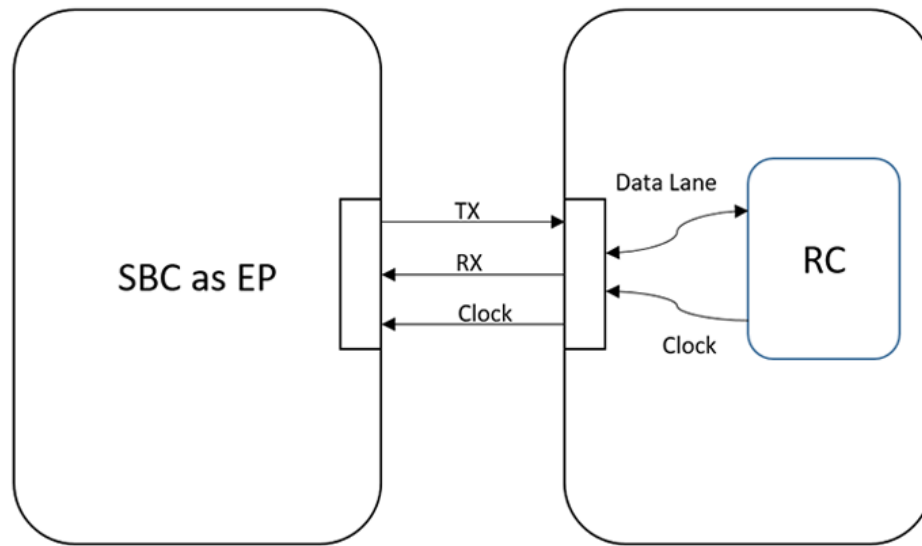


Figure 17: PCIe EP

Table 6: PCIe FPC Interface Pin Definition

Pin	Name	Pin	Name
1	GND (Reference Ground)	21	PCIE_RX0P (Lane 0 Receive Positive)
2	PCIE_RX3N (Lane 3 Receive Negative)	22	GND (Reference Ground)
3	PCIE_RX3P (Lane 3 Receive Positive)	23	PCIE_TX0N (Lane 0 Transmit Negative)
4	GND (Reference Ground)	24	PCIE_TX0P (Lane 0 Transmit Positive)
5	PCIE_TX3N (Lane 3 Transmit Negative)	25	GND (Reference Ground)
6	PCIE_TX3P (Lane 3 Transmit Positive)	26	PCIE_CLKN (External Clock Negative, 100MHz HCSL)
7	GND (Reference Ground)	27	PCIE_CLKP (External Clock Positive, 100MHz HCSL)
8	PCIE_RX2N (Lane 2 Receive Negative)	28	GND (Reference Ground)
9	PCIE_RX2P (Lane 2 Receive Positive)	29	PCIE_FUNC_3V3 (User-Defined Function)
10	GND (Reference Ground)	30	PCIE_PERST_N_3V3 (Reset Signal)
11	PCIE_TX2N (Lane 2 Transmit Negative)	31	PCIE_CLKREQ_N_3V3 (Clock Request Signal)
12	PCIE_TX2P (Lane 2 Transmit Positive)	32	PCIE_WAKE_N_3V3 (Wake-Up Signal)
13	GND (Reference Ground)	33	GND (Reference Ground)
14	PCIE_RX1N (Lane 1 Receive Negative)	34	5.0V (5V Power - Disabled in EP Mode)
15	PCIE_RX1P (Lane 1 Receive Positive)	35	5.0V (5V Power - Disabled in EP Mode)
16	GND (Reference Ground)	36	5.0V (5V Power - Disabled in EP Mode)
17	PCIE_TX1N (Lane 1 Transmit Negative)	37	5.0V (5V Power - Disabled in EP Mode)
18	PCIE_TX1P (Lane 1 Transmit Positive)	38	GND (Reference Ground)
19	GND (Reference Ground)	39	GND (Reference Ground)

Table 6: PCIe FPC Interface Pin Definition

Pin	Name	Pin	Name
20	PCIE_RX0N (Lane 0 Receive Negative)		

4.3.5. MIPI Camera Interface (CSI)

The SBC provides two MIPI CSI interfaces for connecting cameras.

- **CSI01 FPC Interface:** Independent 4-lane camera interface.
- **CSI23 FPC Interface:** Hardware-multiplexed with the MIPI DSI interface; software configuration is required to set it to CSI mode.

Tip

When connecting the FPC cable, the gold fingers should face toward the RJ45 Ethernet port.

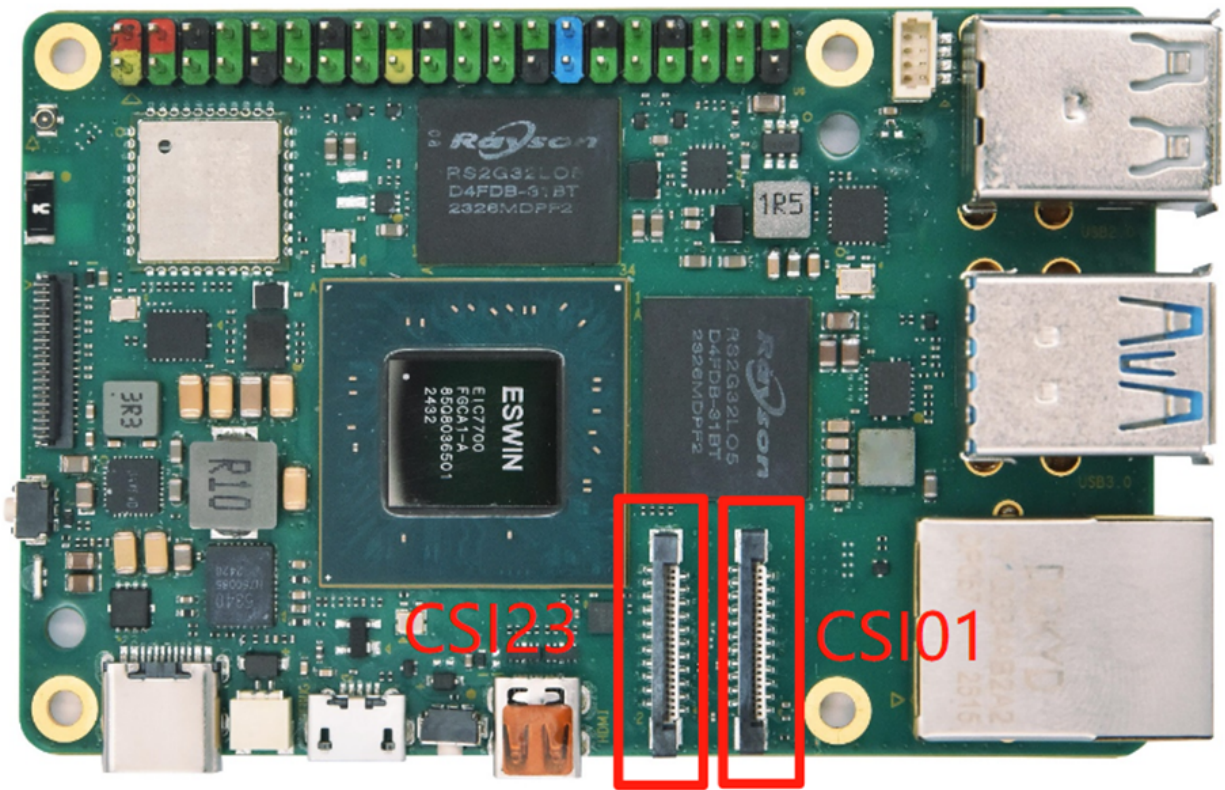


Figure 18: MIPI CSI

Table 7: MIPI CSI01 FPC Interface Pin Definition

Pin No.	Pin Name	Description	Pin No.	Pin Name	Description
1	GND	Reference Ground	12	MIPI_CSI1_D0P	CSI1 Data Lane 0 Receive Positive

Table 7: MIPI CSI01 FPC Interface Pin Definition

Pin No.	Pin Name	Description	Pin No.	Pin Name	Description
2	MIPI_CSI0_D0N	CSI0 Data Lane 0 Receive Negative	13	GND	Reference Ground
3	MIPI_CSI0_D0P	CSI0 Data Lane 0 Receive Positive	14	MIPI_CSI1_D1N	CSI1 Data Lane 1 Receive Negative
4	GND	Reference Ground	15	MIPI_CSI1_D1P	CSI1 Data Lane 1 Receive Positive
5	MIPI_CSI0_D1N	CSI0 Data Lane 1 Receive Negative	16	GND	Reference Ground
6	MIPI_CSI0_D1P	CSI0 Data Lane 1 Receive Positive	17	GPIO_072	General Purpose GPIO, GPIO72, internal pull-up
7	GND	Reference Ground	18	GPIO_070	General Purpose GPIO, GPIO70, internal pull-up
8	MIPI_CSI0_CKN	CSI0 Clock Lane Receive Negative	19	GND	Reference Ground
9	MIPI_CSI0_CKP	CSI0 Clock Lane Receive Positive	20	SCL	I2C3 Clock Line, 3.3V level, internal pull-up
10	GND	Reference Ground	21	SDA	I2C3 Data Line, 3.3V level, internal pull-up
11	MIPI_CSI1_D0N	CSI1 Data Lane 0 Receive Negative	22	3.3V	3.3V Power Supply provided by SBC

Table 8: MIPI CSI23 FPC Interface Pin Definition

Pin No.	Pin Name	Description	Pin No.	Pin Name	Description
1	GND	Reference Ground	12	MIPI_D2P	CSI3 Data Lane 0 Receive Positive
2	MIPI_D0N	CSI2 Data Lane 0 Receive Negative	13	GND	Reference Ground
3	MIPI_D0P	CSI2 Data Lane 0 Receive Positive	14	MIPI_D2N	CSI3 Data Lane 1 Receive Negative
4	GND	Reference Ground	15	MIPI_D2P	CSI3 Data Lane 1 Receive Positive
5	MIPI_D1N	CSI2 Data Lane 1 Receive Negative	16	GND	Reference Ground
6	MIPI_D1P	CSI2 Data Lane 1 Receive Positive	17	GPIO_021	General Purpose GPIO, GPIO72, internal pull-up
7	GND	Reference Ground	18	GPIO_020	General Purpose GPIO, GPIO70, internal pull-up
8	MIPI_CKN	CSI2 Clock Lane Receive Negative	19	GND	Reference Ground
9	MIPI_CKP	CSI2 Clock Lane Receive Positive	20	SCL	I2C2 Clock Line, 3.3V level, internal pull-up
10	GND	Reference Ground	21	SDA	I2C2 Data Line, 3.3V level, internal pull-up
11	MIPI_D2N	CSI3 Data Lane 0 Receive Negative	22	3.3V	3.3V Power Supply provided by SBC

4.3.6. MIPI Display Interface (DSI)

- **Interface Type:** FPC connector with 0.5mm pitch
- **Specification:** 4-lane MIPI DSI TX

Tip

When connecting the FPC cable, the gold fingers shall face toward the RJ45 Ethernet port. This interface is hardware-multiplexed with the MIPI CSI23 interface.

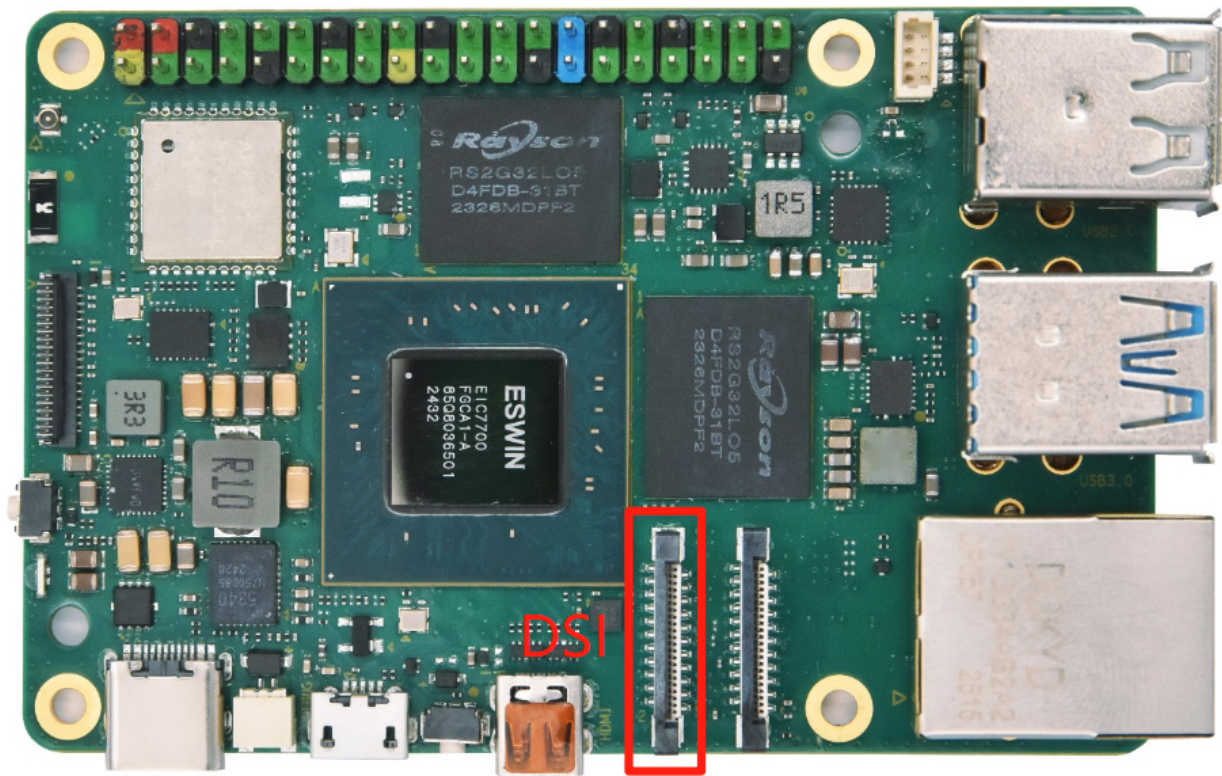


Figure 19: MIPI DSI

Table 9: MIPI DSI FPC Interface Pin Definition

Pin No.	Pin Name	Description	Pin No.	Pin Name	Description
1	GND	Reference Ground	12	MIPI_D2P	DSI Data Lane 0 Transmit Positive
2	MIPI_D0N	DSI Data Lane 0 Transmit Negative	13	GND	Reference Ground
3	MIPI_D0P	DSI Data Lane 0 Transmit Positive	14	MIPI_D2N	DSI Data Lane 1 Transmit Negative
4	GND	Reference Ground	15	MIPI_D2P	DSI Data Lane 1 Transmit Positive
5	MIPI_D1N	DSI Data Lane 1 Transmit Negative	16	GND	Reference Ground

Table 9: MIPI DSI FPC Interface Pin Definition

Pin No.	Pin Name	Description	Pin No.	Pin Name	Description
6	MIPI_D1P	DSI Data Lane 1 Transmit Positive	17	GPIO_021	General Purpose GPIO, GPIO72, internal pull-up
7	GND	Reference Ground	18	GPIO_020	General Purpose GPIO, GPIO70, internal pull-up
8	MIPI_CKN	DSI Clock Lane Transmit Negative	19	GND	Reference Ground
9	MIPI_CKP	DSI Clock Lane Transmit Positive	20	SCL	I2C2 Clock Line, 3.3V level, internal pull-up
10	GND	Reference Ground	21	SDA	I2C2 Data Line, 3.3V level, internal pull-up
11	MIPI_D2N	DSI Data Lane 0 Transmit Negative	22	3.3V	3.3V Power Supply provided by SBC

4.3.7. Fan Interface

- **Interface Type:** 4-Pin 1.0mm connecto
- **ower Supply:** 5V
- **Function:** Supports PWM speed control and tachometer (speed feedback).

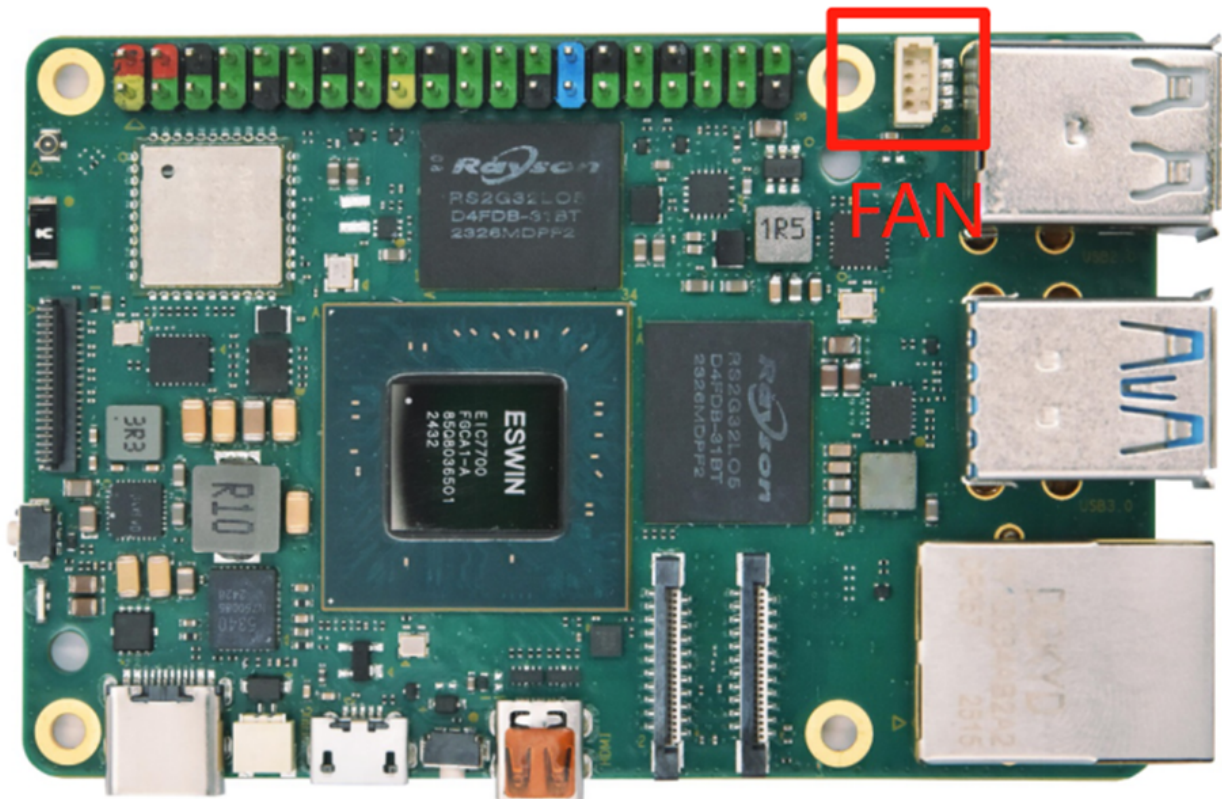


Figure 20: SBC Fan Interface

Table 10: Fan Interface Pin Definition

Pin No.	Pin Name	Description
1	5.0V	5V Power Supply for Fan
2	PWM	PWM Signal for Fan Speed Control
3	GND	Reference Ground
4	TACH	Speed Feedback Signal

4.3.8. RTC Battery Interface

2-Pin 1.0mm connector, used to connect a 3V coin cell battery to maintain the operation of the real-time clock (RTC) when the system is powered off.

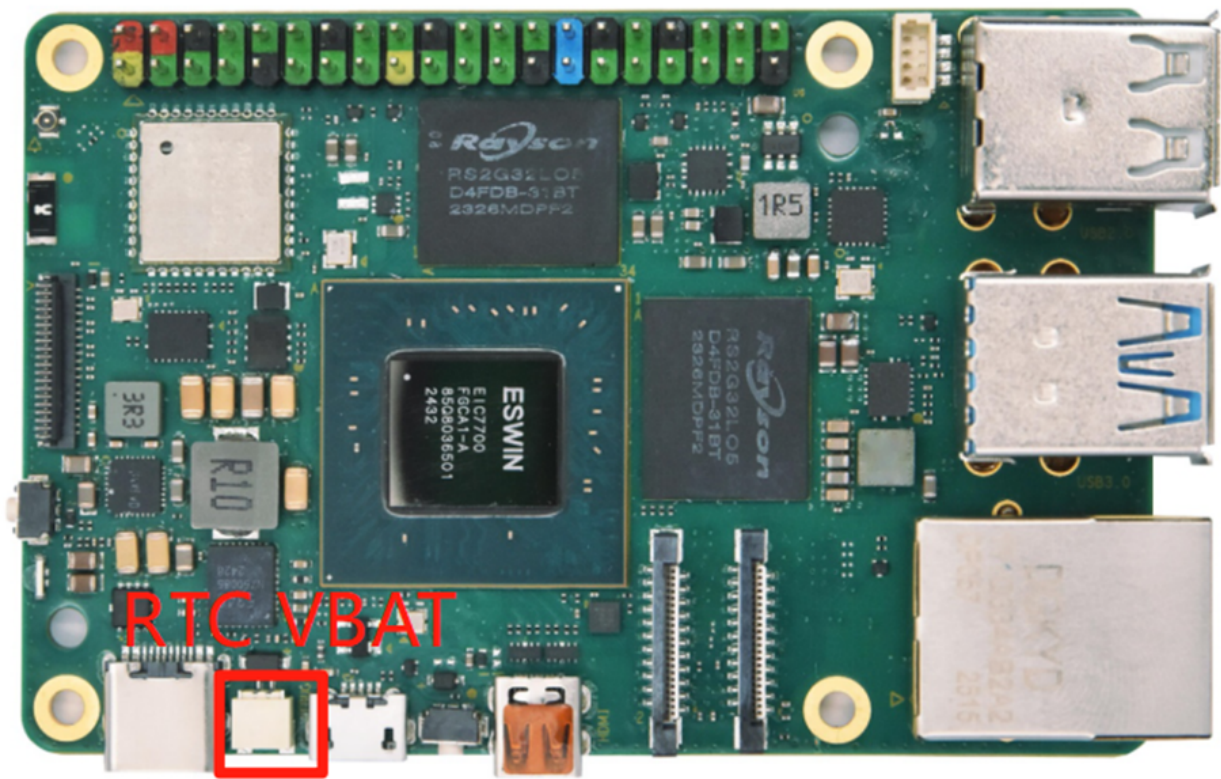


Figure 21: RTC Battery Interface

4.3.9. Wi-Fi Function

- The SBC integrates a 2.4G/5G dual-band Wi-Fi module supporting 802.11 a/b/g/n/ac protocols, connected via the SDIO interface.

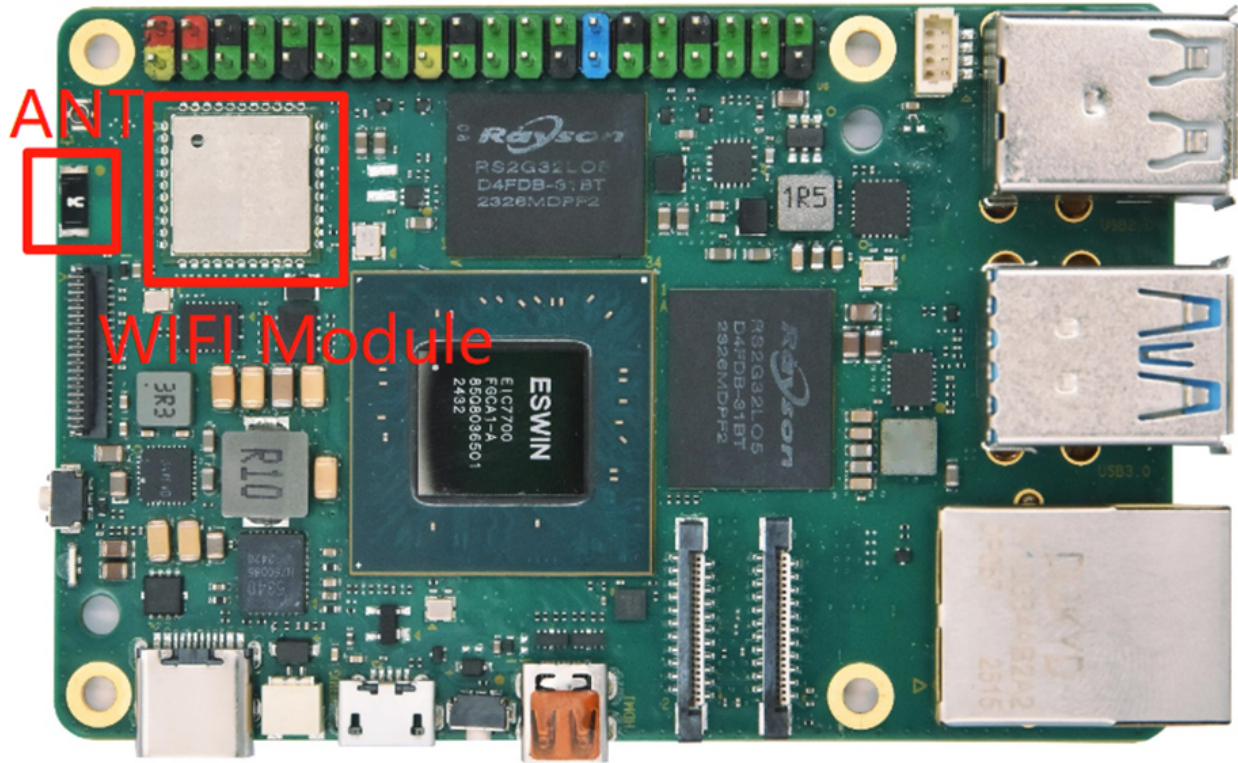


Figure 22: WIFI

- **Antenna:** An onboard antenna is used by default. An external IPEX antenna connector is also reserved. The RF path can be switched by modifying onboard resistors. The modification location is shown in the figure below.

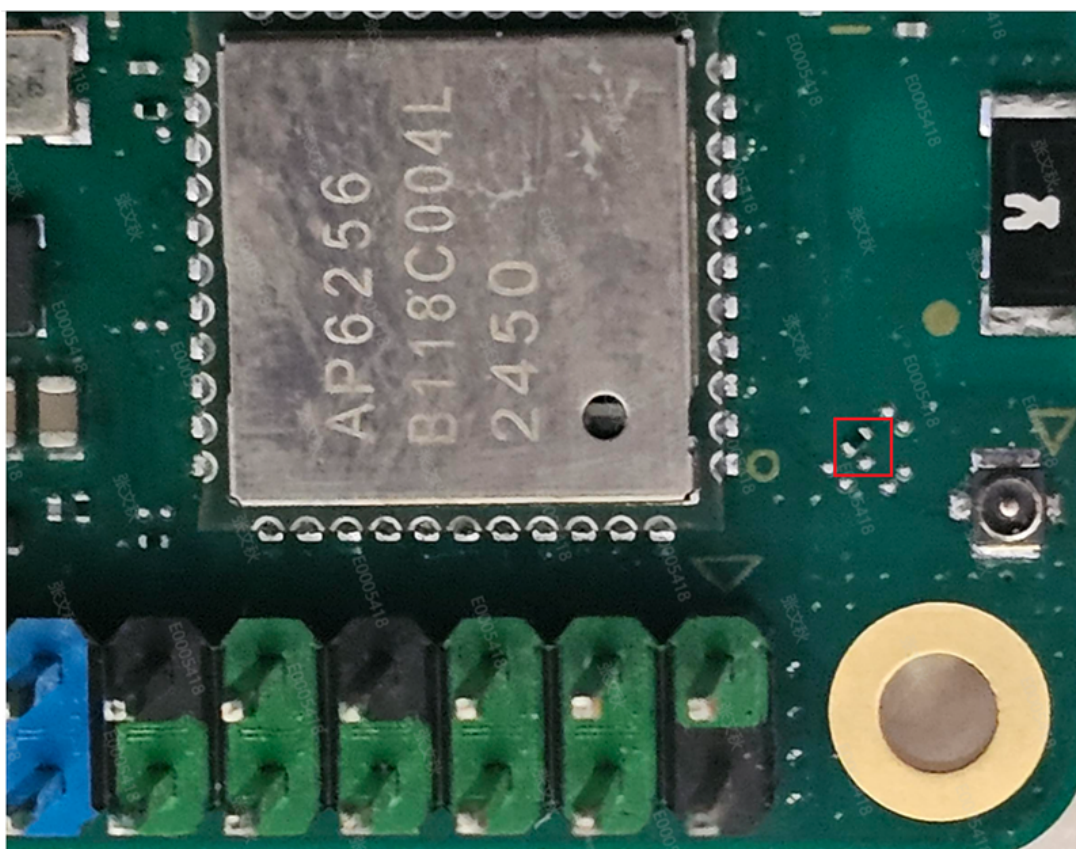


Figure 23: Wi-Fi Resistor Modification

5. System Image Burning Guide

This chapter guides you on how to burn or update the system software for your SBC (Single-Board Computer). The SBC is preloaded with the Ubuntu operating system and adopts a hybrid boot architecture: U-Boot (Bootloader) is stored in the onboard SPI Flash, while the complete Ubuntu system is stored on a removable Micro SD card.

Therefore, the burning operation is divided into two parts:

1. **Bootloader Burning/Recovery:** Targets the onboard SPI Flash and is used for system booting.
2. **Ubuntu System Burning:** Targets the Micro SD card and contains the operating system itself.

Before starting, please make sure you have:

1. Completed all preparations in the [Hardware Introduction and User Guide](#), especially the installation of the heat sink.
2. Installed the necessary serial port drivers according to your development environment (Windows/Linux), such as CH340/CH341 drivers for Windows.
3. Prepared the following cables:
 - **USB Type-C PD power cable** (for power supply)
 - **USB Type-A male-to-male data cable** (for forced recovery mode)
 - **Micro USB to USB Type-A data cable** (for serial port debugging)

The SBC supports updating the Bootloader via USB forced recovery mode and installing the Ubuntu system using an SD card flashing tool.

5.1. Obtaining System Images

ESWIN (Eswin Computing) officially provides pre-built Ubuntu system images for the SBC single-board computer.

1. Download the image: Download the `SBC-xxxx-ubuntu.tar.zst` archive from the [official website](#) or specified resource location to your local device.
2. Extract the image: Use a tool that supports the zstd compression format for extraction (the following example uses the Linux command line):

```
tar --zstd -xvf SBC-xxxx-ubuntu.tar.zst
```

3. Obtain core files: After extraction, you will get an `SBC/` directory containing the following three core files:

File Name	Function Description	Key Purpose & Burning Target
boot-loader_SBC-A1.bin	Main Bootloader (U-Boot) image	The boot program that runs first after system power-on. Used to upgrade an existing usable Bootloader. Burning target: SPI Flash.
recovery_boot-loader_SBC-A1.bin	Bootloader recovery image	Used for system repair when the Bootloader in SPI Flash is corrupted and cannot boot normally. Must be used with Forced Recovery Mode .
sb-ubuntu-24.04-preinstalled-server-riscv64_*.img	Ubuntu full system image	An all-in-one image file containing both the Ubuntu kernel and root filesystem. Burning target: Micro SD card.

Tip

If you need to customize the system for specific requirements (such as kernel configuration, drivers, or software packages), please refer to the separate [BSP Quick Start Guide](#) for system compilation.

Danger**Pre-upgrade Must-Read: Bootloader Pre-upgrade Guide**

Background: The new version EIC7X-26.06 introduces a Bootloader A/B backup mechanism to prevent bricking during upgrades. Due to changes in the partition structure and the old Bootloader's lack of support for the new A/B switching logic, old versions (20251230 and earlier) cannot be directly overwritten using the `es_burn` command. You must manually upgrade the Bootloader first.

Operation Paths (choose one):

1. Option 1 (Recommended): Refer to [Scenario 2: Bootloader fails to boot\(USB Mode Recovery\)](#) and complete the upgrade through the repair process.
2. Option 2 (Manual Tool): Use the dedicated bootloader-tool for command-line flashing.
 - Extract the `bootloader_eic7700-sbc.bin` file from the image package `SBC-26.06-ubuntu.tar.zst`, and download the `bootloader-tool` file from the official website.
 - After booting the old system, transfer these two files to the device system directory (e.g., `/home/eswin/`).
 - Open a terminal, navigate to the file directory (e.g., `cd /home/eswin/`), and execute the following commands in order:

```
sudo ./bootloader-tool getinfo 0
sudo ./bootloader-tool upgrade bootloader_eic7700-sbc.bin
```

- After flashing is complete, please restart the device and enter Uboot mode.
- Once device connection is confirmed normal, you can proceed to upgrade the main system following the standard burning process.

5.2. Bootloader Burning

The Bootloader (U-Boot) is stored in the SPI Flash. Its burning process is divided into two scenarios based on the current system state.

5.2.1. Scenario 1: Bootloader Boots Normally (USB Drive Upgrade)

This method is suitable for regular upgrades where the Bootloader can run and enter the U-Boot command line.

5.2.1.1. Preparations

1. **Prepare a USB drive:** Format a USB drive to the ext4 or FAT32 file system. The ext4 format is recommended to avoid the 4GB size limit that may be encountered when processing large files later.
2. **Place the image file:** Copy the `bootloader_SBC-A1.bin` file to be upgraded to the root directory of the USB drive.

5.2.1.2. Operation Steps

1. **Connection & Boot:** Insert the prepared USB drive into the USB 2.0 Type-A port. Power on the SBC, and press the Enter key during the U-Boot boot countdown to enter the U-Boot command line.



Figure 24: Hardware connection diagram for bootloader flashing

2. **Detect USB Device:** Enter the following command in the U-Boot command line to initialize USB and scan storage devices.

```
=> usb reset
```

After executing the command, confirm that the output contains `1 Storage Device(s) found`. This indicates the USB drive has been successfully recognized. If not found, check that the USB drive is properly inserted, formatted correctly, and connected to the correct USB port.

3. **List USB drive files:** Enter the following command to view the files in the root directory of the USB drive. Confirm that `bootloader_SBC-A1.bin` exists.

```
=> ls usb 0
```

If the USB drive has multiple partitions and the image is not on the first partition, use `ls usb 0:2` to view the second partition (and so on).

4. **Load image to memory:** According to the USB file system, use the corresponding command to load the Bootloader image to the specified memory address (`0x90000000`).

- **ext4 USB drive:**

```
=> ext4load usb 0 0x90000000 bootloader_SBC-A1.bin
```

- **FAT32 USB drive:**

```
=> fatload usb 0 0x90000000 bootloader_SBC-A1.bin
```

After a successful load, the terminal will display the number of bytes read and the transfer speed.

5. **Burn Image to SPI Flash:** Execute the following critical command to write the in-memory image to the onboard SPI Flash.

```
=> es_burn write 0x90000000 flash
```

This process will output detailed erasure and writing progress. Be sure to wait until the final prompt `bootloader write OK` appears, indicating successful burning.

Tip

```
=> es_burn write 0x90000000 flash
Bootspi flash write protection disabled
SF: 4096 bytes @ 0x1000 Read: OK
SF: 4096 bytes @ 0x0 Read: OK
UPDATE BOOTLOADER FAILED (Invalid image file type)
Bootspi flash write protection enabled
exit not allowed from main input shell.
```

If the above error message appears, please erase the flash first and then rewrite it. The operation steps are as follows:

- Erase command: `es_burn erase flash`
- Write command: `es_burn write 0x90000000 flash`

6. **Reboot the System:** After burning is complete, enter the following command to reboot the board. The new Bootloader will take effect immediately.

```
=> reset
```

5.2.2. Scenario 2: Bootloader Fails to Boot (Forced Recovery Mode)

Use this method to repair the system when the Bootloader is corrupted and cannot boot. **This is the core recovery method for the SBC.**

5.2.2.1. Forced Recovery Mode

Use this mode for repair when the Bootloader in SPI Flash is corrupted and the system cannot boot. The SBC provides two ways to enter this mode. You can enter Forced Recovery Mode using either of the following methods (Method 1 is recommended):

1. **Connect to host then power on (Recommended):** With the SBC fully powered off, connect its onboard USB 3.2 Type-A port to a PC using a USB cable, then power on the board. For details, see [Forced Recovery Operation Steps](#).
2. **Hold recovery key then power on:** With the SBC fully powered off, press and hold the recovery key `S2`, then power on the board. Release `S2`, after the onboard LED lights up. For key location, see [Button Functions and Operation](#) below.

5.2.2.2. Preparations

Hardware Connections:

1. Use a USB Type-A male-to-male cable to connect its onboard USB 3.2 Type-A port (the bottom one) to your PC.
2. Use a Micro USB cable to connect the SBC debug serial port to your PC for log viewing.
3. Connect the Type-C PD power supply, but do not power on yet.

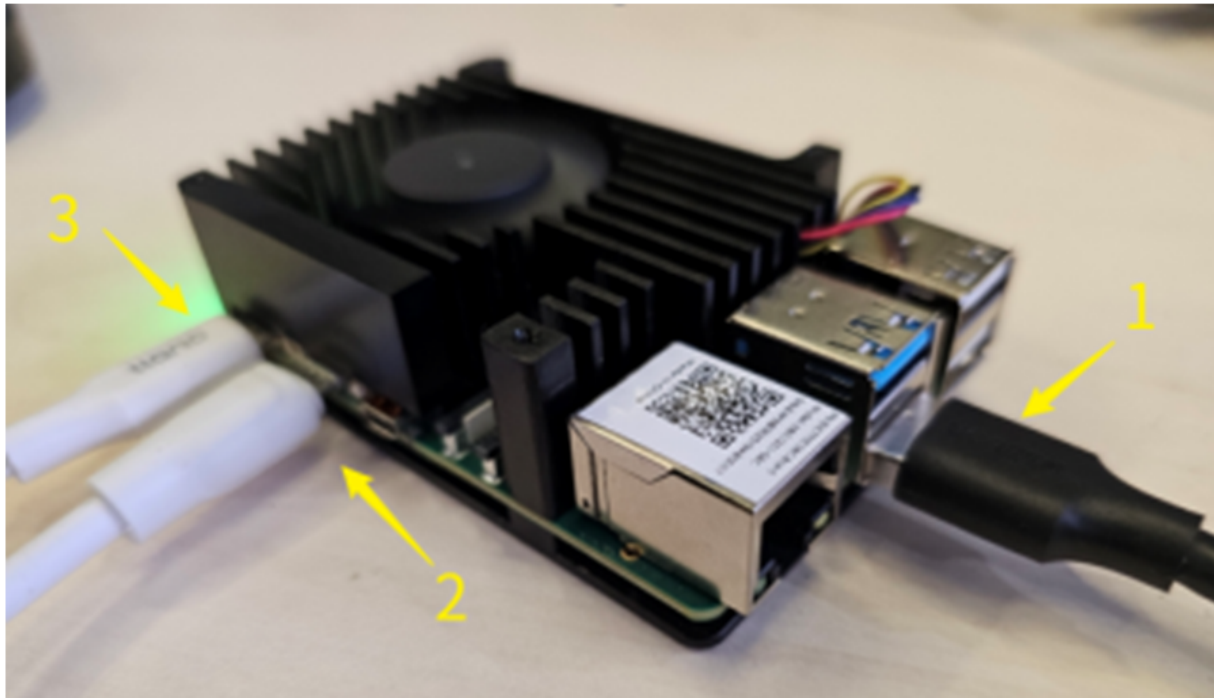


Figure 25: Recover Hardware Connection Diagram

Prepare the image: Prepare the `recovery_bootloader_SBC.bin` file on the PC.

5.2.2.3. Forced Recovery Operation Steps

1. **Enter Forced Recovery Mode:** With the SBC fully powered off, connect the USB cable to the PC first, then power on the SBC. The PC will recognize a removable disk named `ESWIN-2030`.
2. **Copy the recovery image:** Copy or drag the `recovery_bootloader_SBC.bin` file into the `ESWIN-2030` drive.
3. **Wait for automatic burning:** After the file is copied, the SBC will start burning automatically. You can check the progress via the serial terminal; the success indicator is the message `bootloader write OK with logo` in the log.
4. **Finish recovery:** After burning, power off the board, disconnect the USB cable from the PC, then power on again to boot normally.

5.3. Burn Ubuntu System Image

The Ubuntu system for the SBC must be burned to a Micro SD card.

5.3.1. Preparations

1. **Prepare a USB drive:** Prepare a USB drive with a capacity of at least 16 GB, and format it to the ext4 file system.
2. **Obtain the image file:** Download and extract the Ubuntu system package from [Obtaining Images](#) to get the file `sbc-ubuntu-24.04-preinstalled-server-riscv64_*.img` (abbreviated as `ubuntu-*.img` below).
3. **Place the image:** Copy the full-disk `ubuntu-*.img` image file to the root directory of the USB drive.

5.3.2. Operation Steps

Please follow the steps below strictly in order.

1. **Connect USB drive and boot:** Insert the prepared USB drive into the USB 2.0 Type-A port. Power on the SBC, and press Enter during the U-Boot boot countdown to enter the U-Boot command line.



Figure 26: Hardware connection diagram for ubuntu flashing

2. **Initialize and recognize the USB drive:** Enter the following command in the U-Boot command line to scan for USB storage devices:

```
=> usb reset
```

After executing the command, confirm that the output includes `1 Storage Device(s) found`. This means the USB drive has been recognized successfully. If not found, check if the USB drive is properly connected, formatted correctly, and inserted into the right USB port.

3. **Verify the image file:** Enter the following command to list the files in the root directory of the USB drive and confirm the Ubuntu full-disk image file exists:

```
=> ls usb 0
```

If the USB drive has multiple partitions and the image file is not on the first partition, use `ls usb 0:2` to check the second partition (and so on).

4. **Burn the full-disk image:** Run the `es_fs write` command to write the image file to the target storage device byte by byte, automatically overwriting all existing partitions and data.

```
=> es_fs write usb 0 ubuntu-*.img mmc 0
```

Tip

- The full exact filename is required for the burn command; the wildcard `*` cannot be used. The filename above is for demonstration purposes only.
- If you cannot copy the full name completely, run the command `setenv stdout serial` first, then copy and paste.
- This process will take a long time (depending on image size and USB drive speed). Do NOT power off or perform any operations until the command finishes.

5. **Completion and Reboot:** After the burn command finishes successfully, enter the following command to reboot the board:

```
=> reset
```

The board will boot from the newly burned Ubuntu system.

5.3.2.1. First Login & Password

After logging into the system for the first time, you will be required to change the default password. Depending on your operational environment, you can follow either of the methods below:

Method 1: Change Password via the System Login Interface

1. Connect an HDMI display and USB keyboard, system boot up and enter the login interface.
2. Use the default username in the login window.
3. Enter the default password: ubuntu (in lowercase).
4. After logging in, a prompt will appear below the password input box: You are required to change your password immediately (administrator enforced)
5. As prompted, enter the current password again: ubuntu.
6. Enter a new password (the new password cannot be the same as the old one) and retype it to confirm.
7. After successfully changing the password, the system will proceed to the desktop as normal.

Method 2: Change Password via Serial Port Login

1. Connect to the device using a serial port tool and access the login interface.
2. Enter the username: ubuntu.
3. Enter the default password: ubuntu.
4. The system will prompt you to change the password: You are required to change your password immediately (administrator enforced)
5. As prompted, enter the current password: ubuntu.
6. Set a new password and enter it again to confirm (the new password cannot be the same as the old one).
7. Upon successful modification, the system will display the welcome interface: Welcome to Ubuntu

```
ubuntu login: ubuntu # Enter username ubuntu
Password:          # Enter password ubuntu
You are required to change your password immediately (administrator enforced).
Changing password for ubuntu.
Current password:   # Enter password again ubuntu
New password:      # Enter new password
Retype new password: # Re-enter new password
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.6.115-2025-eic7700 riscv64)
```

Tip

Special scenario: If you are using an Ubuntu image pre-installed or modified by ESWIN, the default password may have been changed. In this case, try logging in with username `ubuntu` and password `Eswin888888`.

6. Frequently Asked Questions (FAQ)

This chapter summarizes common issues you may encounter when using the SBC and their solutions. Please refer to it before use or when problems occur.

6.1. System Boot & U-Boot

Q1: How to familiarize myself with U-Boot commands and functions?

A: At the U-Boot command line, you can get help at any time using the following methods:

- Enter `help` or `?` to view a list of all available commands.
- Enter `help <command_name>` to view detailed usage instructions for a specific command (e.g., `help mmc`).

Q2: What should I do if characters are truncated or lost when pasting long commands via the serial port?

A: This is a default limitation of the U-Boot serial input buffer. The issue can be resolved by modifying the environment variables:

```
=> setenv stdout serial
=> saveenv
```

Tip

This setting will make all U-Boot output appear only on the serial terminal, and no longer output to the HDMI display. To restore HDMI output, run `setenv stdout vidconsole, serial` followed by `saveenv`.

Q3: Does U-Boot support USB hot-plugging?

A: Yes. After inserting a USB flash drive, run the `usb reset` command in the U-Boot command line to rescan and recognize the device.

Q4: Does U-Boot support SD card hot-plugging?

A: No. The SD card must be inserted before powering on the system; otherwise, it cannot be recognized.

6.2. Network Connection

Q5: What should I do if the development board has no network connection, cannot ping successfully, or fails to obtain an IP address?

A: Please troubleshoot in the following order:

1. **Physical Connection:** Ensure the Ethernet cable is securely connected to the development board (SoC Gigabit Ethernet port) and the router/switch.
2. **MAC Address Check:** Run `print ethaddr` in the U-Boot command line to verify that the MAC address is properly configured. The output should be in the format `ethaddr=xx:xx:xx:xx:xx:xx`.
3. **Set MAC Address:** If `ethaddr` is empty or needs to be modified, run the following commands (replace the example address with a valid MAC address):

```
=> env default -a
=> env set ethaddr aa:bb:cc:11:22:33
=> saveenv
=> reset
```

6.3. System Flashing and Partitioning

Q6: The system reports errors or fails to boot after flashing.

A: This is usually due to incompatibility between the old U-Boot environment variables and the new system. After flashing the new Bootloader, try resetting the environment variables using the following steps:

```
=> env default -a
=> saveenv
=> reset
```

Tip

`env default -a` clears all manually configured environment variables (such as MAC address, boot parameters, etc.). Please use with caution and reconfigure after execution.

Q7: How to verify that the flashed system image (Bootloader/Kernel) has been updated successfully?

A: Check the serial log during system boot:

- **Bootloader:** At the early stage of U-Boot startup, a line shows the U-Boot build date and time. Compare this time with the build time of the image you flashed.
- **Linux Kernel:** During kernel decompression and startup, its build timestamp is also printed. Comparing this time confirms whether the kernel image has been updated.

6.4. Display Output

Q8: How to switch display output from HDMI to MIPI DSI?

A: Follow the steps below:

1. First, when the power is disconnected, you need to connect the raspberry 7-inch screen with the MIPI DSI interface. After startup, log in to the ubuntu system.

2. Operate the following command:

- `config /boot/overlay.txt`

When `enable_overlay=1`, a new overlay is merged;

when `enable_overlay=0`, it reverts to the original state.

```
root@ubuntu:/usr/bin# vim /boot/overlay.txt
#dtb overlay

enable_overlay = 1
dtbo=/boot/dtbs/6.6.115-2025-eic7700/eswin/overlays/eic7700-sbc-a1-mipi.dtbo
~
```

- `config /usr/bin/merge_dtbo`

```
#!/bin/bash
# apply dtbo overlay
set -e
CONFIG_FILE="/boot/overlay.txt"
KERNEL_VERSION="$(uname -r)"
LINK_DTB="dtb-${KERNEL_VERSION}"
BASE_DTB_DIR="/boot/dtbs/${KERNEL_VERSION}/eswin"
BASE_DTB="eic7700-sbc-a1.dtb"
MAIN_DTB="${BASE_DTB_DIR}/${BASE_DTB}"
OVERLAY_DIR="${BASE_DTB_DIR}/overlays"
MERGE_DTB="eic7x-merged.dtb"
OUTPUT_DTB="/boot/${MERGE_DTB}"
APPLY_DTB="${BASE_DTB_DIR}/${MERGE_DTB}"
```

- run ./merge_dtbo

```
root@ubuntu:/usr/bin# ./merge_dtbo
Parse overlay file: /boot/overlay.txt
Add overlay dtbo: /boot/dtbs/6.6.115-2025-eic7700/eswin/overlays/eic7700-sbc-a1-mipi.dtbo
Parse overlay.txt done: find 1 dtbo
  1. /boot/dtbs/6.6.115-2025-eic7700/eswin/overlays/eic7700-sbc-a1-mipi.dtbo
enable_overlay = 1
=== Applying overlay ..... ===
Applying overlay, merge 1 dtbo
Start merging 1 overlay dtbo to base dtb
main dtb /boot/dtbs/6.6.115-2025-eic7700/eswin/eic7700-sbc-a1.dtb
Apply overlay: /boot/dtbs/6.6.115-2025-eic7700/eswin/overlays/eic7700-sbc-a1-mipi.dtbo
  ✓ success apply: /boot/dtbs/6.6.115-2025-eic7700/eswin/overlays/eic7700-sbc-a1-mipi.dtbo
Overlay merge done
Verifying DTB file...
DTB verify pass !
Overlay is applied
```

- After completing the command and restarting, you can use the MIPI DSI interface to output.

To switch back to HDMI, the user needs to config enable_overlay=0, and run ./merge_dtbo then restart, it's OK.

Tip

The specific DTB file name in the /boot/dtbs/ directory may vary by system version. Please refer to the actual file name.